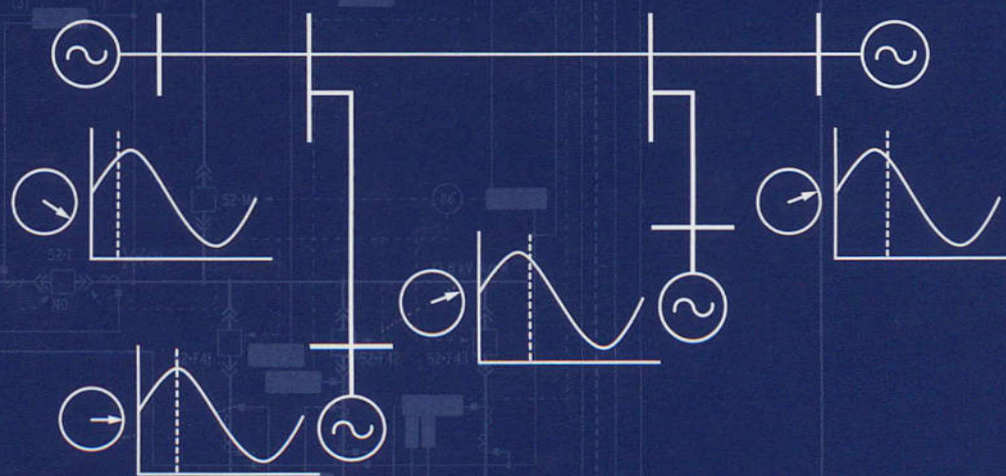


Modern Solutions

for Protection, Control, and Monitoring of

Electric Power Systems



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3 Distribution System Protection, Automation, and Monitoring

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3.1 Introduction

Most consumers get electric power from distribution systems. Because these systems generally lack redundancy, many faults result in loss of service to customers.

Relays, reclosers, and fuses with time-coordinated characteristics that respond to phase and residual currents protect most distribution systems. Sequencing and reclosing schemes of relays and reclosers use instantaneous elements, which are generally difficult to coordinate. Schemes that use instantaneous elements must balance the pros and cons of sensitivity, fuse-saving, trip-saving, customer service, cost, and complexity.

Customer demand for greater reliability, plus an increase in distributed generation, will encourage the design of distribution systems with more redundancy and much faster protection and control.

This chapter explains many techniques that can provide faster tripping, greater sensitivity, and improved security. We also explain how to meet greater demands and expectations for service reliability and how to easily accept sources (solar, wind, backup plants), as well as loads, on the same feeder.

This chapter covers:

- Applying negative-sequence overcurrent elements to provide faster, more sensitive protection for phase faults and to reduce tradeoffs between phase-fault sensitivity and load current.
- Improving ground fault protection sensitivity in distribution systems.
- Using load-encroachment elements to prevent phase overcurrent protection from operating on load.
- Applying modern multifunction relays to protect distribution systems with distributed generation.
- Taking advantage of the communication built into today's relays for high-speed local and distributed communication-based schemes.
- Using fiber-optic light sensors to detect arc-flash events and provide high-speed fault clearing.
- Applying modern relays and recloser controls to automate distribution networks.

- Using fault location from modern relays and fault-path information from faulted circuit indicators to promptly pinpoint the faulted section and expedite line repair.

3.2 Limitations of Traditional Overcurrent Protection

Most of today's distribution feeders are radial and are protected by overcurrent devices, including relays, reclosers, and fuses. However, traditional overcurrent protection has many limitations:

- Phase overcurrent relays must be set above the maximum load current. Therefore, maximum load expectations limit the sensitivity and speed of the protection for phase-to-phase and three-phase faults.
- Protection settings must be checked against load levels frequently and seasonally, because load growth and extreme temperatures can result in load current tripping unfaulted circuits when energy is most needed.
- Ground overcurrent relays must be set above the maximum load unbalance expected on the feeder. Unbalance, whether from uneven loading of the three phases, from single-phase switching operations, or from blown fuses, limits the sensitivity of ground protection.

Overcurrent protection coordinates by current, time, or both. Figure 3.1 depicts a radial system with two inverse-time overcurrent devices. To ensure selectivity, the upstream (backup) device must wait for the downstream (primary) device to clear the fault. Otherwise, customers between the upstream and downstream devices will lose service unnecessarily. The time-current curve of the upstream device must be above the downstream device curve, with some time margin. Unfortunately, this level of selectivity requires a sacrifice in speed. Fault-clearing time increases as the fault moves to the line sections located closer to the source, where the fault current is higher. The undesirable result is longer clearing times for more severe faults.

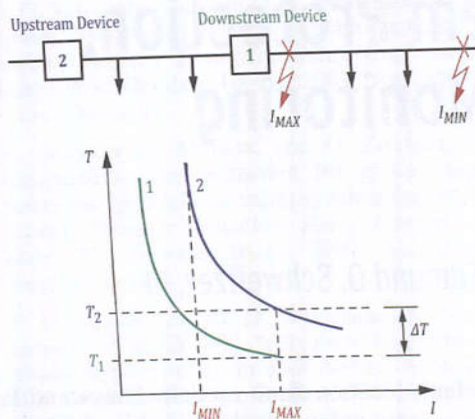


Figure 3.1 Overcurrent protection is time-delayed to provide coordination.

Although instantaneous overcurrent elements provide faster tripping, they can be coordinated only when the feeder impedance creates substantial difference in fault current between close-in and remote faults. In tightly knit distribution systems, such as those in urban areas, in manufacturing plants, and aboard ships, these elements become difficult or impossible to coordinate, because the fault current may not change significantly from the source to the farthest load.

Delayed fault clearing risks safety and equipment damage. Also, some loads are sensitive to voltage sags caused by faults. Delayed tripping may even initiate voltage collapse. In addition, delayed tripping could cause transient stability problems.

Fortunately, there are many new techniques to get beyond these and other limitations of traditional phase and ground overcurrent protection.

3.3 Modern Solutions for Distribution System Protection, Automation, and Monitoring

Today, microprocessor-based relays provide many advanced functions, and economical radio channels are available. Now, at low cost, you can apply the protection, control, and monitoring solutions traditional in transmission systems to distribution systems.

3.3.1 New abilities

Microprocessor-based relays have new abilities that significantly improve overcurrent protection. The SEL-351S, shown in Figure 3.2, is one example. In addition to the traditional overcurrent elements, this relay includes voltage and directional elements, communications abilities, and many control and monitoring functions. This functionality allows protection of complex distribution systems, such as looped networks and systems with distributed generation sources (Section 3.8). SEL microprocessor-based recloser controls

(Figure 3.3) also provide these functions. Installed at sectionalizing points of the distribution system, these controls improve protection and provide fast automatic network reconfiguration. The SEL-751A also provides arc-flash protection for switchgear installations to prevent safety hazards for personnel. This relay uses fiber-optic light sensors to detect arc-flash events and initiate fast fault clearing (Section 3.10).

3.3.2 More sensitive fault detection

Microprocessor-based relays provide negative-sequence overcurrent elements (Section 3.4), as well as the traditional phase and ground overcurrent elements. Because negative-sequence overcurrent elements can be set below load current, they provide protection for phase-to-phase faults that is almost as sensitive as the protection ground elements provide for ground faults.

If complemented with negative-sequence overcurrent elements, phase overcurrent elements are required only for three-phase fault coverage. Torque-controlling phase overcurrent elements with load-encroachment elements (Section 3.7) reduces their susceptibility to operating on load. Load-encroachment elements discriminate between load and three-phase faults by measuring the angle of the positive-sequence impedance. Of course, phase overcurrent elements back up the negative-sequence overcurrent elements.

The sensitivity gained using negative-sequence and load-encroachment elements is particularly valuable on long feeders (low end-of-line fault current) and on the backup relays that sense the total current from the transformer into all the feeders (high load current).

Highly sensitive elements available in SEL relays detect ground faults on ungrounded and high-impedance grounded systems. An arc-sensing element available in the SEL-451 relay detects high-impedance ground faults on multigrounded distribution circuits. Section 3.6 describes these elements.

3.3.3 Faster fault clearing

Modern relays provide a variety of overcurrent elements. By combining instantaneous elements and short-time, definite-time elements with inverse-time elements, you can improve coordination, thus reducing fault-clearing times without sacrificing security. Negative-sequence overcurrent elements and arc-flash protection elements further improve speed.

The low cost of modern transformer and bus protection relays makes high-speed differential protection applicable in many distribution substations. In addition, modern relays, when used with communication, provide:

- Zone-interlocked protection (Section 6.4.5), an alternative to bus differential protection that uses information from feeder relays to discriminate bus faults from feeder faults.
- Breaker-failure protection (Section 6.5) using transformer and feeder relays for faster backup.
- High-speed distribution and subtransmission line protection using fiber-optic or spread-spectrum radio channels (Section 3.9).

3.3.4 Faster service restoration

Modern relays and recloser controls can speed up restoration of service via throw-over schemes and fast network reconfiguration schemes (Section 3.11). The group settings available in modern relays and recloser controls accommodate configuration changes in the distribution system. You can design protection systems that automatically select the settings group required for each system configuration.

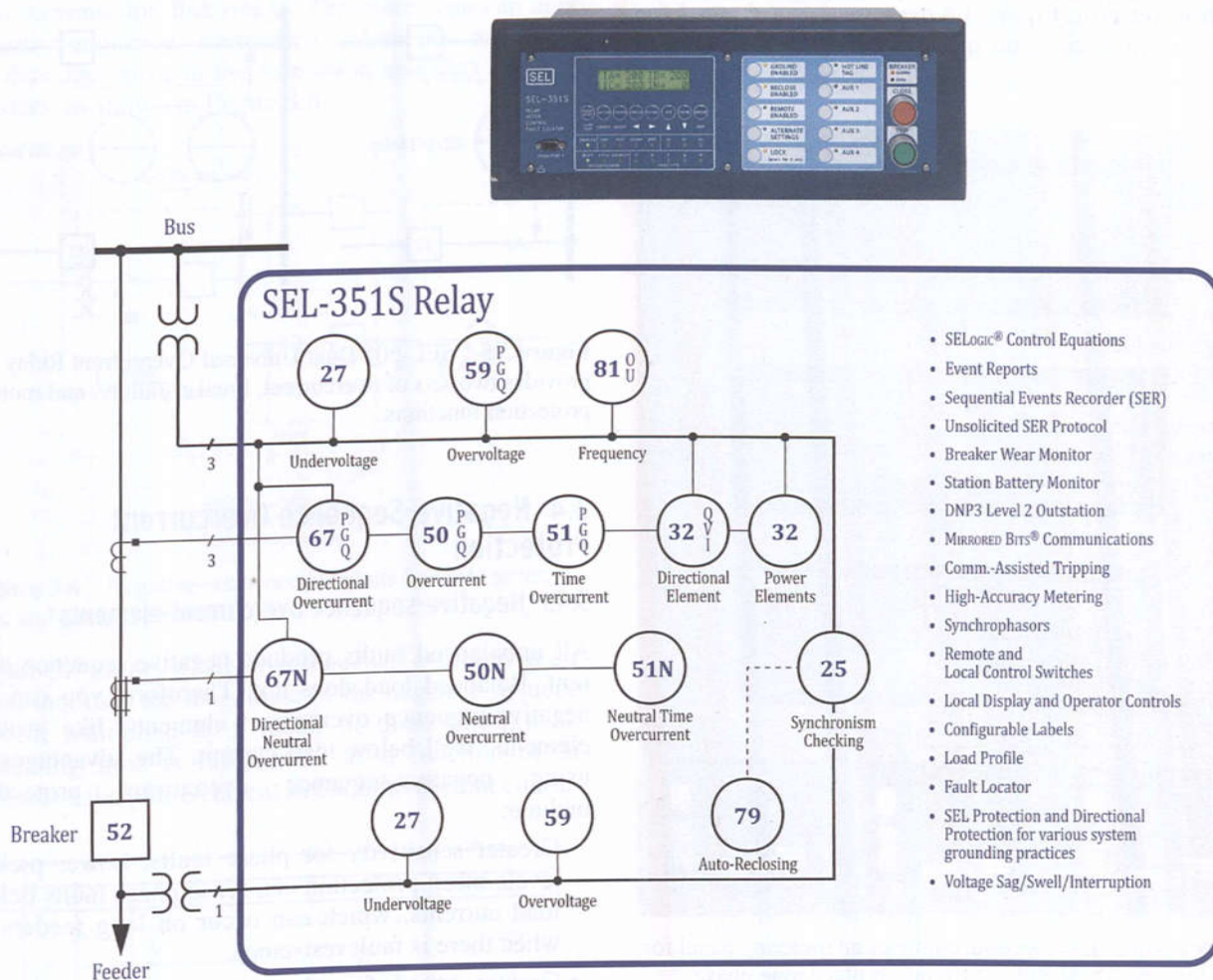


Figure 3.2 Functional diagram of the SEL-351S Protection System.



Figure 3.3 Microprocessor-based multifunction recloser controls. (a) SEL-351R Recloser Control. (b) SEL-651R Advanced Recloser Control.

Combining faulted circuit indicators with the fault-locating abilities of modern relays reduces the time needed to determine the faulted line section and provide information to line repair crews (Section 3.12).

3.3.5 Higher reliability and lower cost

Replacing many electromechanical relays with one microprocessor-based relay saves panel space and reduces wiring, providing much higher reliability at much lower cost. Figure 3.4 depicts part of a panel for protection, control, and metering of three distribution feeders.

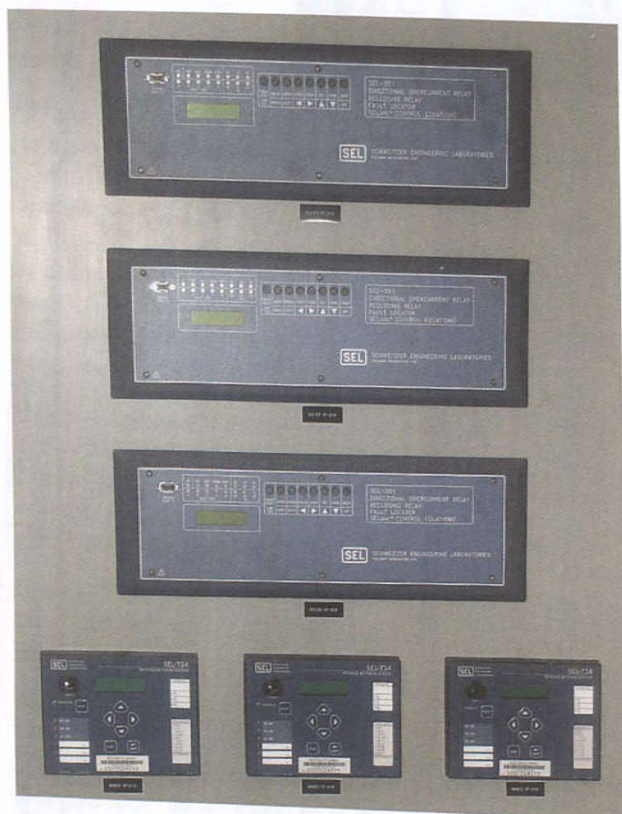


Figure 3.4 SEL protection, control, and metering panel for three feeders. Three SEL-351 relays replace nine phase relays, three ground relays, three reclosing relays, and many auxiliary relays.

Microprocessor-based relays also self-test. When the relay detects an abnormal hardware or software condition, it disables critical functionality, such as protection, and closes a contact as notification that service is required. Wired to the communications network, this contact improves monitoring of relay status. Relay diagnostic ability increases reliability and reduces cost, because it almost eliminates the need for routine maintenance (Section 12.6 and [1]).

New solutions include a dual relay that provides two independent groups of protection functions. One relay provides very low-cost overcurrent protection for two

feeders, as shown in Figure 3.5(a). This relay can also provide breaker-failure protection. Two relays (Figure 3.5(b)) provide full redundant protection for two feeders. Because the SEL-501 relay also includes dual motor protection functions, it can protect two motors or a feeder and a motor. Another solution, given the low cost of modern relays, is to use two relays per circuit, such as a pair of SEL-351 relays, for excellent reliability. Chapter 12 discusses some possibilities.

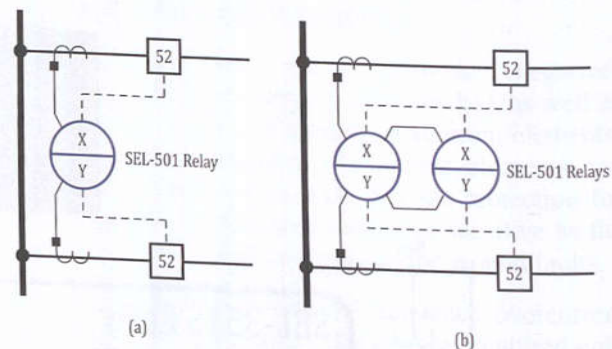


Figure 3.5 SEL-501 Dual Universal Overcurrent Relay provides two sets of overcurrent, breaker-failure, and motor protection functions.

3.4 Negative-Sequence Overcurrent Protection

3.4.1 Negative-sequence overcurrent elements

All unbalanced faults produce negative-sequence current. Balanced load does not. Therefore, you can set negative-sequence overcurrent elements, like ground elements, well below load currents. The advantages of using negative-sequence overcurrent protection include:

- Greater sensitivity for phase faults. Lower pickup levels mean protection can sense phase faults below load currents, which can occur on long feeders or when there is fault resistance.
- Greater speed for phase faults. Figure 3.10 and Figure 3.11 show the speed advantage, as well as the sensitivity advantage.
- Backup for ground faults. An open neutral path in the current transformer (CT) secondaries disables ground overcurrent elements. However, the negative-sequence overcurrent elements still function, so they provide ground fault protection.
- Backup through delta-wye transformers. Ground overcurrent elements cannot detect ground faults through delta-wye transformers, but negative-sequence overcurrent elements do detect these faults.
- Fast and sensitive feeder backup protection at the bus.

- Standard in most modern relays: nothing extra to buy, wire, or maintain. SEL introduced negative-sequence overcurrent elements more than 15 years ago.
- Easy to understand, coordinate, and set.

Negative-sequence overcurrent elements (50Q/51Q) with instantaneous, definite-time, and inverse-time characteristics are available in SEL distribution relays and recloser controls, as well as in transformer, bus, and transmission line relays. Therefore, you can apply negative-sequence overcurrent elements to protect feeders and buses in the substation and also along the feeders, as shown in Figure 3.6.

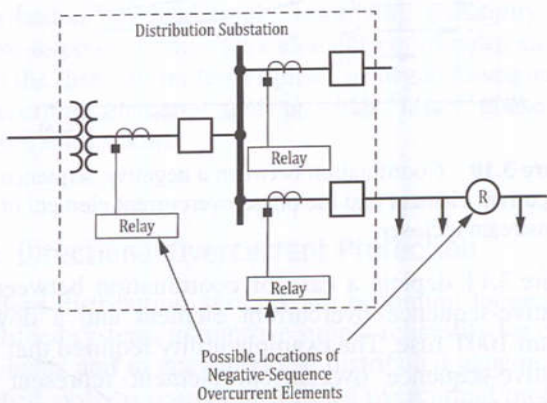


Figure 3.6 Negative-sequence elements benefit feeder, bus, and backup protection.

Figure 3.7 shows a negative-sequence overcurrent element that uses the magnitude of the negative-sequence current multiplied by a factor of three [2], [3], [4]. Operating time is represented as a function of $3I_2$ instead of the phase current in the time-current curve.

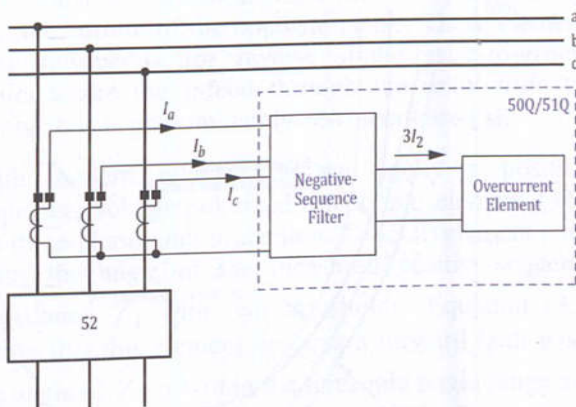


Figure 3.7 Negative-sequence overcurrent element.

3.4.2 Coordinating negative-sequence overcurrent elements with phase overcurrent elements

Negative-sequence overcurrent elements work alongside phase overcurrent relays, reclosers, and fuses (Figure 3.8).

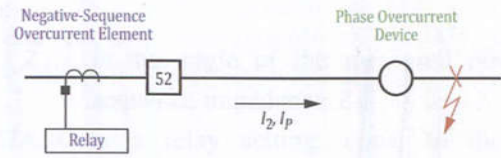


Figure 3.8 Negative-sequence overcurrent elements easily coordinate with devices that respond to phase currents.

Fault analyses reported in [2] and [4] compare the values of $3I_2$ and phase current I_P for different types of faults on the radial distribution system shown in Figure 3.8. Regardless of the fault type, $3I_2$ is never larger than $\sqrt{3}$ times the phase current I_P

$$3I_2 \leq \sqrt{3}I_P \quad (3.1)$$

A negative-sequence overcurrent element with its pickup set to $\sqrt{3}$ times the pickup of a phase overcurrent device will have equal or less sensitivity for all fault types than the phase device. As a consequence, these devices will maintain time coordination for all fault types, as long as they have similar curve shapes.

For coordination purposes, we define an equivalent phase overcurrent element of the negative-sequence overcurrent element. We can apply the normal procedure to coordinate this equivalent phase overcurrent element with the downstream phase device. After selecting the pickup, curve type, and time dial of the equivalent phase overcurrent element, we calculate the true pickup setting of the negative-sequence overcurrent element using (3.2).

$$(3I_2)_{PU} = \sqrt{3}I_{EP} \quad (3.2)$$

where:

$(3I_2)_{PU}$ is the negative-sequence overcurrent element pickup setting;

I_{EP} is the equivalent phase overcurrent element pickup resulting from the coordination process.

Figure 3.9 shows the time-current curves of the negative-sequence overcurrent element and its equivalent phase element. For coordination, we only use the equivalent phase element curve. Current I_{EP} may be set below load current. References [2] and [4] provide the guidelines for coordinating negative-sequence overcurrent elements.

If the downstream protective device includes ground overcurrent elements, in addition to phase overcurrent elements, there is no need to check coordination between the negative-sequence overcurrent element and the ground overcurrent element of the downstream device. The phase and ground overcurrent elements of the downstream device will always operate faster than the negative-sequence overcurrent element for all types of faults.

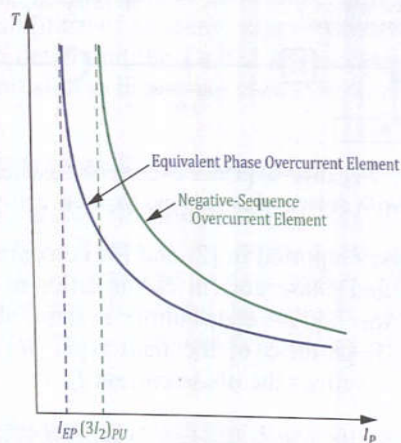


Figure 3.9 Defining an equivalent phase element, you can use the normal coordination procedure for negative-sequence overcurrent elements.

Ground overcurrent elements fail to operate for ground faults if the neutral conductor of the CT secondary circuit is open, because no residual current flows. Negative-sequence overcurrent elements provide some degree of backup for ground overcurrent elements within the same multifunction relay. For a single-phase-to-ground fault, the fault current (referred to CT secondary) flows on the phase conductor of the faulted-phase CT. Because the neutral conductor is open, this current returns through the other two phase conductors. As a result, the negative-sequence overcurrent elements measure approximately 1.5 times the true negative-sequence current for a single-phase-to-ground fault. Thus, these negative-sequence overcurrent elements (measuring higher than the true negative-sequence current value) may not coordinate with a load-side protective device if a ground fault occurs beyond the load-side device.

Figure 3.10 depicts an example of coordination between a negative-sequence overcurrent element and a downstream line recloser. The maximum load current through the feeder relay is 450 A, and the maximum load current through the recloser is 150 A. We set the recloser phase element pickup to $(I_R)_{PU} = 200$ A to accommodate load and cold-load pickup current. We can set the equivalent phase element below the maximum load current of 450 A. In this example, we select $I_{EP} = 300$ A. Based on (3.2), we set $(3I_2)_{PU} = 520$ A.

Figure 3.10 also shows a traditional phase element set to coordinate with the recloser. The 450-ampere feeder load current limits the sensitivity of the feeder phase element to a pickup of $(I_P)_{PU} = 600$ A. The negative-sequence overcurrent element backs up the recloser for faults above 300 A and is much faster than the phase element for faults above 600 A.

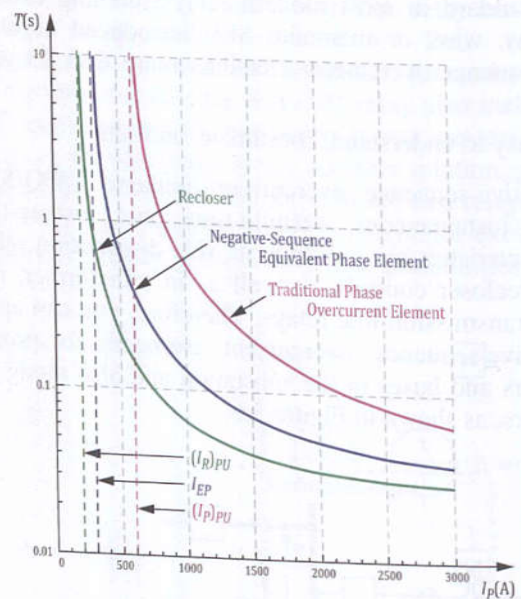


Figure 3.10 Coordination between a negative-sequence overcurrent element and the phase overcurrent element of a downstream recloser.

Figure 3.11 depicts a case of coordination between a negative-sequence overcurrent element and a downstream 100T fuse. The example utility required that the negative-sequence overcurrent element represent an improvement over the traditional phase overcurrent element, which is set at 600-amperes pickup, Very Inverse characteristic, and Time Dial = 2, to coordinate with the fuse. The 100T fuse melts at about 250 A. To achieve pickup coordination, the equivalent phase element should be set at about $I_{EP} = 375$ A. To achieve time coordination with the fuse, the utility chose a Very

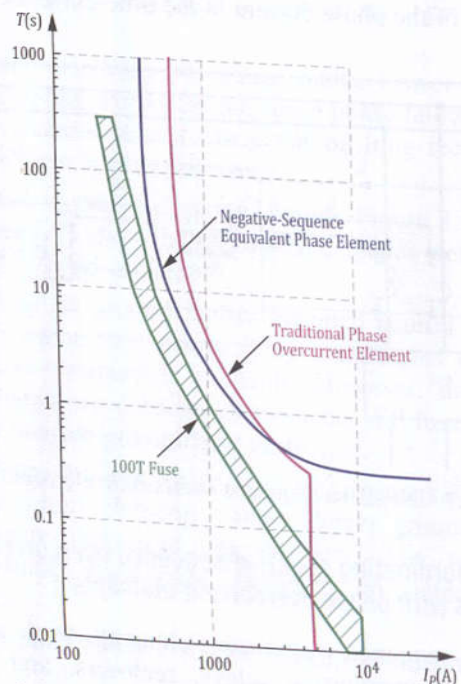


Figure 3.11 Coordination between a negative-sequence overcurrent element and a downstream 100T fuse (courtesy of BC Hydro).

Inverse characteristic and set Time Dial = 3.3. Based on (3.2), the pickup setting of the negative-sequence overcurrent element is $(3I_2)_{PU} = 650$ A. Figure 3.11 shows the major improvement in sensitivity and speed the negative-sequence overcurrent element provides.

Applying a negative-sequence overcurrent element to distribution bus overcurrent protection further improves sensitivity and speed. Traditional phase overcurrent elements must be set above the combined load of all the feeders on the bus, plus any emergency load conditions. Such settings greatly reduce sensitivity and speed for bus protection and feeder backup protection (Section 3.7). Pickup for the negative-sequence overcurrent element, however, can be significantly below distribution bus load levels, improving sensitivity for phase-to-phase faults. This element coordinates easily with the distribution feeder phase or negative-sequence overcurrent elements and provides faster phase-to-phase fault backup.

3.5 Directional Overcurrent Protection

Modern distribution systems are becoming looped or multifeed systems to improve supply reliability for critical loads and to accommodate distributed generation. Looped systems require directional overcurrent protection (67/67N), where a directional element controls the overcurrent element operation.

3.5.1 Directional elements for phase fault protection

Traditional directional elements for phase fault protection typically have a 90-degree connection. This connection requires three directional elements. Each directional element measures a phase-to-phase voltage and the current of the opposite phase. These elements may misoperate for reverse single-phase-to-ground faults where the infeed through the relay from the remote bus is predominantly zero sequence [5].

With modern relays, you can apply a positive-sequence, voltage-polarized directional element (32P) for three-phase fault protection. The 32P element compares the angle of the measured positive-sequence impedance $\bar{Z}_1$ with two thresholds. Equation (3.3) shows that this element declares a forward fault when the angle of $\bar{Z}_1$ is within the threshold angle range and declares a reverse fault when the angle of $\bar{Z}_1$ is outside the angle range.

$$(-90^\circ + ZIANG) < \text{Arg} \left[\bar{Z}_1 \right] < (90^\circ + ZIANG) \quad (3.3)$$

(Forward Fault)

where:

$\text{Arg} \left[\bar{Z}_1 \right]$ is the angle of the measured positive-sequence impedance $\bar{Z}_1$;

$ZIANG$ is a relay setting, equal to the line positive-sequence impedance angle.

Figure 3.12 shows the operating characteristic of the 32P element.

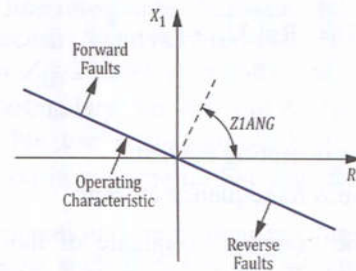


Figure 3.12 The SEL positive-sequence, voltage-polarized directional element (32P) provides directional discrimination for three-phase faults.

We enable the 32P element when all three phase-to-phase current magnitudes (I_{ab} , I_{bc} , and I_{ca}) are above a set threshold and the 32Q element is disabled.

Phase-to-phase and phase-to-phase-to-ground faults produce negative-sequence currents and voltages. You should apply a negative-sequence, voltage-polarized directional element (32Q), a directional element responding to the negative-sequence impedance, to detect these faults (Section 3.5.2.2). The 32P element combines with the 32Q element to provide directional supervision for all phase faults. The 32Q element has priority over the 32P element.

3.5.2 Directional elements for ground fault protection

Ground directional elements can respond to negative- or zero-sequence quantities. Traditional ground directional elements require users to select the polarizing and operating quantities for each application. The relay uses these quantities at all times, a restriction that may cause the directional element to misoperate when system configuration changes.

SEL invented a ground directional element [6] that selects the best polarizing and operating quantities for each ground fault, based on system conditions. Applying this new ground directional element eliminates the need to make choices and compromises. Another benefit is that this element does not require user settings.

This SEL ground element, applicable to effectively and low-impedance grounded systems, includes three directional elements: zero-sequence, current-polarized (32I); negative-sequence, voltage-polarized (32Q); and zero-sequence, voltage-polarized (32V) directional elements. The negative- and zero-sequence, voltage-polar-

ized directional elements respond to impedance, solving the voltage sensitivity problem of traditional voltage-polarized elements for remote or high-impedance ground faults [5].

3.5.2.1 Zero-sequence, current-polarized directional element

The 32I element uses (3.4) to calculate the scalar quantity T .

$$T = \text{Re} \left[3\bar{I}_0 \cdot \bar{I}_{POL}^* \right] \quad (3.4)$$

where:

$3\bar{I}_0$ is the operating current;

$\bar{I}_0$ is the zero-sequence current;

$\bar{I}_{POL}^*$ is the complex conjugate of the polarizing current $\bar{I}_{POL}$.

The 32I element compares T with two thresholds. When T is positive and above the positive threshold, the element declares a forward ground fault. When T is negative and below the negative threshold, the element declares a reverse ground fault.

All directional elements require supervision. The relay enables the 32I element when the zero-sequence current I_0 is greater than the positive-sequence current I_1 times a preset a_0 factor ($I_0 > a_0 I_1$). The a_0 factor increases the 32I element security for zero-sequence currents that circulate during three-phase faults because of line asymmetries or CT saturation [6].

3.5.2.2 Negative-sequence, voltage-polarized directional element

The 32Q element serves as a ground directional element. The 32Q element calculates the scalar quantity z_2 using (3.5) [5], [6].

$$z_2 = \frac{\text{Re} \left[\bar{V}_2 \left(\bar{I}_2 \cdot 1 \angle ZIANG \right)^* \right]}{I_2^2} \quad (3.5)$$

where:

$\bar{V}_2$ is the negative-sequence voltage;

$\bar{I}_2$ is the negative-sequence current;

I_2 is the magnitude of $\bar{I}_2$;

$ZIANG$ is a relay setting, equal to the line positive-sequence impedance angle.

The 32Q element compares z_2 with two thresholds. If z_2 is below a forward-fault threshold, the 32Q element

declares a forward fault. If z_2 is above a reverse-fault threshold, the 32Q element declares a reverse fault.

The relay enables the 32Q element when the negative-sequence current I_2 is greater than the positive-sequence current I_1 times a preset a_2 factor ($I_2 > a_2 I_1$) and the negative-sequence current I_2 is greater than the zero-sequence current I_0 times a preset k factor ($I_2 > k I_0$). When the second condition is not true, the relay selects the 32V element. This check ensures that the relay uses the most robust sequence quantities. The a_2 factor increases the 32Q element security in the same way that the a_0 factor increases the 32I and 32V element security.

3.5.2.3 Zero-sequence, voltage-polarized directional element

The 32V element is the zero-sequence analogy of the 32Q element. The 32V element uses equation (3.6) to calculate the scalar quantity z_0 .

$$z_0 = \frac{\text{Re} \left[3\bar{V}_0 \left(3\bar{I}_0 \cdot 1 \angle ZOMTA \right)^* \right]}{(3I_0)^2} \quad (3.6)$$

where:

$\bar{V}_0$ is the zero-sequence voltage;

$\bar{I}_0$ is the zero-sequence current;

I_0 is the magnitude of $\bar{I}_0$;

$ZOMTA$ is the maximum torque angle of the 32V element (a relay setting).

The 32V element makes directional decisions in the same way as the 32Q element. The element compares z_0 against the forward-fault and reverse-fault thresholds to determine the direction of the ground fault.

The relay enables the 32V element when $I_0 > a_0 I_1$ and I_2 is not greater than $k I_0$. Zero-sequence currents resulting from line asymmetries are normally smaller than negative-sequence currents for most phase conductor configurations [7]. Therefore, the a_0 factor is usually smaller than the a_2 factor.

3.5.2.4 32Q and 32V element operation for ground faults

Figure 3.13 shows the negative-sequence network for a ground fault at the remote terminal of the protected line in a two-source system. In this network:

- $\bar{Z}_{2S}$ and $\bar{Z}_{2R}$ are the negative-sequence impedances of the local and remote sources.
- $\bar{Z}_{2L}$ is the negative-sequence impedance of the protected line.

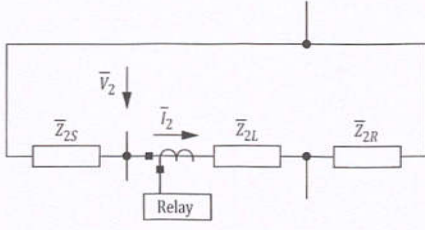


Figure 3.13 Negative-sequence network for a ground fault at the end of the protected line in a two-source system.

The 32Q element measures the negative-sequence voltage $\bar{V}_2$ and the negative-sequence current $\bar{I}_2$. For this forward fault:

- At the relay location, $\bar{V}_2 = -\bar{I}_2 \cdot \bar{Z}_{2S}$.
- If the source negative-sequence impedance angle $\text{Arg}[\bar{Z}_{2S}]$ equals the setting angle ZIANG, the calculated scalar quantity is $z_2 = -|\bar{Z}_{2S}|$.

For any forward-fault location, the directional element measures the negative-sequence impedance of the equivalent system behind the relay.

A similar analysis shows that, for all reverse ground faults, the scalar quantity is $z_2 = |\bar{Z}_{2L} + \bar{Z}_{2R}|$, corresponding to the negative-sequence impedance of the equivalent system in front of the relay.

The 32V element operation for ground faults is similar to that of the 32Q element.

Figure 3.14 shows the operating characteristic of the 32Q (Figure 3.14(a)) and 32V (Figure 3.14(b)) elements and the calculated z_2 and z_0 values for forward and reverse faults. For simplicity, we consider the power system to be purely inductive and assume the maximum torque angle (ZIANG and ZOMTA, respectively) set to 90 degrees. For the 32Q element, we use $\bar{Z}_{1L}$ instead of $\bar{Z}_{2L}$ in Figure 3.14(a). Recall that $\bar{Z}_{1L} = \bar{Z}_{2L}$ for lines.

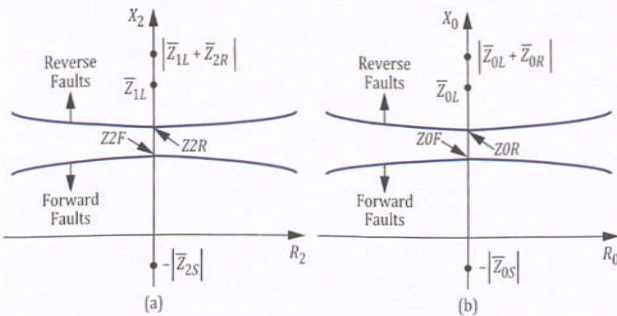


Figure 3.14 Operating characteristics of the SEL 32Q and 32V directional elements.

Because the z_2 (or z_0) values for forward and reverse faults are of opposite signs, there is ample margin for

setting the forward-fault and reverse-fault thresholds. The smallest separation between these values occurs when the sources at both line ends are infinite ($\bar{Z}_{2S} = \bar{Z}_{2R} = 0$, or $\bar{Z}_{0S} = \bar{Z}_{0R} = 0$). For forward faults $z_2 = 0$ (or $z_0 = 0$) and for reverse faults $z_2 = |\bar{Z}_{2L}|$ (or $z_0 = |\bar{Z}_{0L}|$). In most applications, you can set the forward-fault (Z2F, Z0F) and reverse-fault (Z2R, Z0R) thresholds at approximately one-half of the corresponding sequence line impedance. Separate these thresholds to create a security band between them. Possible settings are $Z2F = Z_{1L}/2$ ohms secondary and $Z2R = Z_{1L}/2 + 0.1$ ohms secondary. Set Z0F and Z0R the same as Z2F and Z2R, but use zero-sequence line impedance Z_{0L} instead of positive-sequence line impedance Z_{1L} .

Some applications may require threshold settings closer to zero, such as $Z2F = Z0F = 0$ and $Z2R = Z0R = 0.2$ ohms secondary, or $Z2F = Z0F = -0.3$ and $Z2R = Z0R = 0.3$ ohms secondary. These applications include short lines, parallel lines with single-pole tripping, very strong or weak sources, applications involving transformers, applications where the line impedance is unknown, and applications where overcurrent elements are set very sensitively. You should conduct fault studies to validate these settings [8].

3.5.2.5 Selecting the optimal directional element

Each of the three directional elements (32I, 32Q, 32V) has advantages and disadvantages for various system conditions. In SEL relays, SEL's Best Choice Ground Directional Element[®] logic selects the optimal directional element for a particular fault and system configuration, based on the processing sequence selected by the user and on the directional elements that are already enabled [6].

For example, assume that the user sets the element processing sequence as 32I first, 32Q next, and 32V last. During a fault, the Best Choice Ground Directional Element logic operates as follows:

- The relay uses the 32I element if it is enabled, which requires I_0 and I_{POL} to be above the sensitivity thresholds.
- If the 32I element is not enabled, the relay uses the 32Q element.
- If the 32Q element is not enabled, the relay uses the 32V element, if the 32V element is enabled.

In the automatic setting mode, the SEL ground directional element provides settings for the 32Q and 32V directional elements, simplifying the relay setting procedure. The relay selects the most suitable settings from given system parameters. For example, the relay uses the line impedance parameters to define the Z2F, Z2R, Z0F, and Z0R settings. The relay automatic mode sets the a_0 and a_2 factors to 0.1. These conservative set-

tings make the directional elements secure for almost all line configurations.

3.6 Improving Ground Fault Protection Sensitivity

Ground-fault current depends mainly on the system grounding and the fault resistance. System grounding is an important consideration when selecting the ground fault protection method.

Ground faults in ungrounded, resonant-grounded, and high-resistance grounded systems do not affect the phase-to-phase voltages. Because these systems may operate in a faulted condition, ground relays typically issue an alarm signal instead of tripping the faulted line. These systems require high relay sensitivity because the fault current is very low. Some SEL relays include a highly sensitive, neutral-channel current input and apply sensitive ground fault detection algorithms. SEL Application Guide AG98-10 provides guidelines for setting sensitive ground overcurrent elements. SEL Application Guide AG99-02 discusses the application of sensitive ground directional elements.

Because ground faults in effectively and low-impedance grounded systems may cause high currents, ground relays must trip the faulted element. Detecting high-impedance faults on these systems requires dedicated, high-impedance fault detection elements and sensitive directional elements.

3.6.1 Ungrounded systems

Ungrounded or isolated neutral systems have no intentional connection to ground. For these systems, the distributed capacitance of feeders and other system components provides the path for ground-fault current. In a faulted ungrounded system, the currents of healthy feeders and the faulted feeder current are not significantly different. Overcurrent protection cannot reliably determine the faulted line. A traditional approach is to use a zero-sequence overvoltage relay to detect the ground fault, then manually open each feeder to locate the faulted line. However, this approach greatly increases fault location times and causes interruptions of healthy lines. These problems are unacceptable in present-day distribution systems.

Instead, you can apply the zero-sequence, voltage-polarized directional element for ungrounded systems (32U) [9]. This element uses (3.6) with $ZOMTA = -90^\circ$ to calculate z_0 , and it compares z_0 to forward-fault and reverse-fault thresholds to determine the direction of the ground fault.

Figure 3.15 shows the zero-sequence network for a forward ground fault in an ungrounded system. XC_{0L} is the zero-sequence capacitive reactance of the protected

line, and XC_{0S} is the zero-sequence capacitive reactance of the remaining system. At the relay location, $\bar{V}_0 = -\bar{I}_0 (-jXC_{0S}) = jXC_{0S} \bar{I}_0$. Therefore, the measured zero-sequence impedance (the $\bar{V}_0/\bar{I}_0$ ratio) equals the equivalent capacitive reactance behind the relay $-(-jXC_{0S}) = jXC_{0S}$. A similar analysis shows that the impedance measured for a reverse ground fault equals the protected line capacitive reactance $-jXC_{0L}$ (assuming that $XC_{0L} \gg Z_{0L}$).

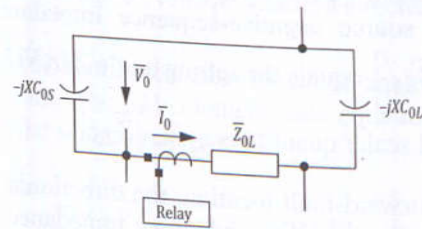


Figure 3.15 Zero-sequence network for a forward ground fault in an ungrounded system.

Figure 3.16(a) shows the phasor diagram for forward and reverse faults. Figure 3.16(b) shows the 32U element operating characteristic. Setting $ZOMTA = -90^\circ$ results in the forward-fault region being on the first and second quadrants of the impedance plane and the reverse-fault region being on the third and fourth quadrants of the impedance plane. The forward-fault and reverse-fault thresholds are set for jXC_{0S} to be in the forward-fault region and $-jXC_{0L}$ to be in the reverse-fault region.

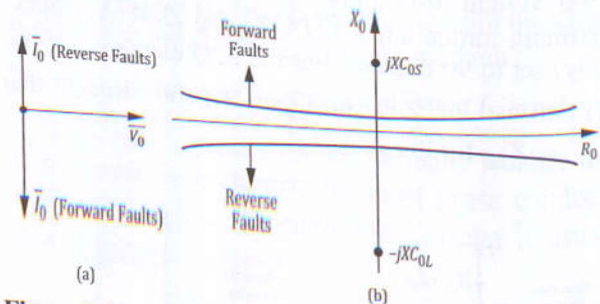


Figure 3.16 The SEL zero-sequence, voltage-polarized directional element for ungrounded systems (32U) provides sensitive and selective ground fault detection.

SEL-351, SEL-351A, and SEL-351S relays, which include the 32U element, have a highly sensitive, neutral-channel current input. Flux-summing CTs are the best choice for ungrounded systems. They are free from the spurious zero-sequence secondary currents that result from phase CT errors, and their CT ratio can be sized properly for high-impedance fault detection.

3.6 Improving Ground Fault Protection Sensitivity

SEL Application Guide AG2002-06 provides guidelines for applying SEL-351 relays to protect ungrounded systems. Reference [10] outlines a method for faulted-phase selection in ungrounded systems, and [11] presents an actual application example.

3.6.2 Resonant-grounded systems

Resonant-grounded or compensated systems are grounded through a variable reactor tuned to the system phase-to-ground capacitance. This high-impedance reactor, referred to as a ground fault neutralizer or a Petersen Coil, is connected to the distribution transformer neutral or a grounding transformer.

A zero-sequence overvoltage relay detects ground faults in resonant-grounded systems. However, the system operator has to disconnect each feeder of the bus to find the faulted line. Wattmetric and incremental conductance elements can provide selective and sensitive ground fault detection. SEL-351, SEL-351A, and SEL-351S relays have wattmetric and incremental conductance elements. These relays also provide highly sensitive, neutral-current inputs because ground-fault currents are quite low in resonant-grounded systems.

3.6.2.1 Wattmetric directional element

Figure 3.17(a) shows the single-line diagram of a resonant-grounded, radial distribution system. The diagram shows the protected line and an equivalent line that represents all the remaining feeders. Figure 3.17(b) shows the system zero-sequence network. C_{0L} and R_{0L} are the protected line zero-sequence capacitance and leakage resistance. C_{0S} and R_{0S} are the corresponding values for the remaining system. L_N and R_N represent the Petersen Coil inductance and resistance.

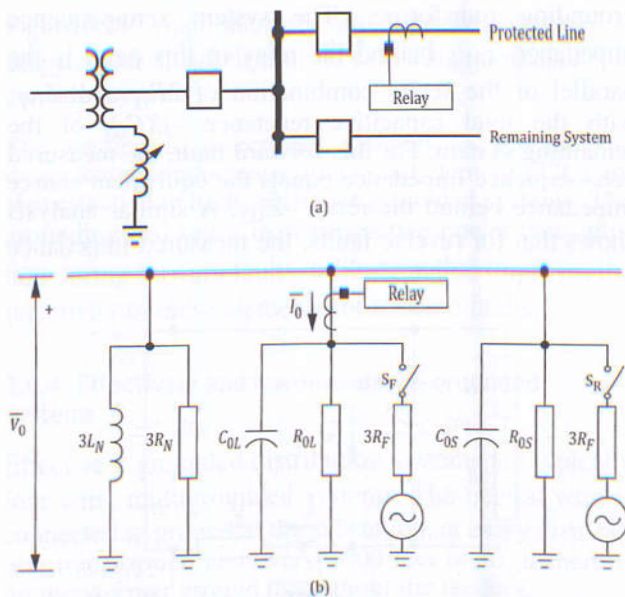


Figure 3.17 Single-line diagram and zero-sequence network of a resonant-grounded distribution system.

We represent the ground fault by connecting an equivalent Thevenin source in series with a resistance (equal to three times the fault resistance R_F) at the fault point. Equation (3.7) gives the relay current for a forward fault (Switch S_F closed and Switch S_R open in Figure 3.17(b)).

$$\bar{I}_0 = -\bar{V}_0 \left(\left(\frac{1}{R_{0S}} + \frac{1}{3R_N} \right) + j \left(\omega C_{0S} - \frac{1}{3\omega L_N} \right) \right) \quad (3.7)$$

(Forward Fault)

Equation (3.8) gives the relay current for a reverse fault (Switch S_F open and Switch S_R closed).

$$\bar{I}_0 = \bar{V}_0 \left(\frac{1}{R_{0L}} + j\omega C_{0L} \right) \quad (\text{Reverse Fault}) \quad (3.8)$$

Figure 3.18(a) shows the phasor diagram for ground faults in resonant-grounded networks. The in-phase (active) component of $\bar{I}_0$ is always positive for reverse faults and negative for forward faults (see (3.7) and (3.8)).

The wattmetric directional element (32W) uses (3.9) to calculate the scalar quantity W_0 [9].

$$W_0 = \text{Re} \left[3\bar{V}_0 \cdot 3\bar{I}_0^* \right] \quad (3.9)$$

where:

$\bar{V}_0$ is the zero-sequence voltage;

$\bar{I}_0^*$ is the complex conjugate of the zero-sequence current $\bar{I}_0$.

The 32W element compares W_0 with forward-fault (negative) and reverse-fault (positive) thresholds. If W_0 is below the forward-fault threshold, the element declares a forward fault. If W_0 is above the reverse-fault threshold, the element declares a reverse fault. Figure 3.18(b) shows the 32W element operating characteristic on the complex power plane.

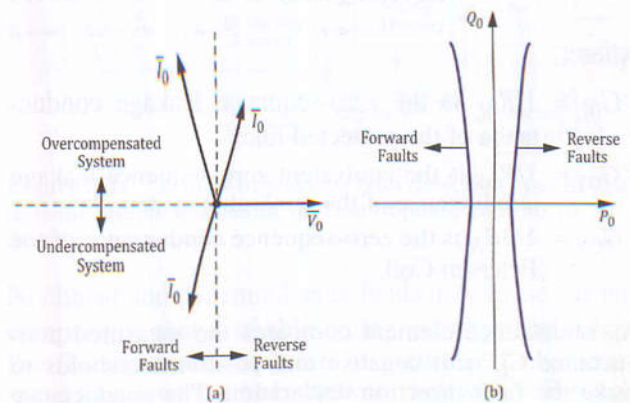


Figure 3.18 SEL wattmetric directional element (32W) provides sensitive and selective ground fault detection in resonant-grounded systems.

A zero-sequence overvoltage element supervises the wattmetric element to provide security. The pickup voltage must be above the maximum value of V_0 for normal system unbalances. A typical setting is 20 percent of the rated system voltage. The overvoltage element sensitivity determines the wattmetric element sensitivity.

The wattmetric method provides ground fault detection in all types of distribution systems that have low ground-fault-current values, including ungrounded, resonant-grounded, and high-resistance grounded systems. Using flux-summing CTs avoids phase CT error problems.

3.6.2.2 Incremental conductance element

According to (3.7) and (3.8), the zero-sequence admittance $\bar{Y}_0$ measured by the relay for forward and reverse faults is as shown in (3.10) and (3.11).

$$\bar{Y}_0 = \frac{\bar{I}_0}{\bar{V}_0} = -\left(\left(\frac{1}{R_{0S}} + \frac{1}{3R_N}\right) + j\left(\omega C_{0S} - \frac{1}{3\omega L_N}\right)\right) \quad (\text{Forward Fault}) \quad (3.10)$$

$$\bar{Y}_0 = \frac{\bar{I}_0}{\bar{V}_0} = \frac{1}{R_{0L}} + j\omega C_{0L} \quad (\text{Reverse Fault}) \quad (3.11)$$

Taking the real component from (3.10) and (3.11), we obtain the conductance G_0 measured by the relay.

$$G_0 = \text{Re}\left[\frac{\bar{I}_0}{\bar{V}_0}\right] = -\left(\frac{1}{R_{0S}} + \frac{1}{3R_N}\right) = -(G_{0S} + G_{0N}) \quad (\text{Forward Fault}) \quad (3.12)$$

$$G_0 = \text{Re}\left[\frac{\bar{I}_0}{\bar{V}_0}\right] = \frac{1}{R_{0L}} = G_{0L} \quad (\text{Reverse Fault}) \quad (3.13)$$

where:

$G_{0L} = 1/R_{0L}$ is the zero-sequence leakage conductance of the protected line;

$G_{0S} = 1/R_{0S}$ is the equivalent zero-sequence leakage conductance of the remaining system;

$G_{0N} = 1/3R_N$ is the zero-sequence conductance of the Petersen Coil.

A conductance element compares the measured conductance G_0 with negative and positive thresholds to make the fault direction declaration. The conductance method works well in balanced systems and for low CT

and CT unbalances introduce errors in the measured conductance and thereby limit sensitivity.

SEL developed the incremental conductance directional element (32C) [9], which uses the incremental zero-sequence current $\Delta\bar{I}_0$ and voltage $\Delta\bar{V}_0$ to calculate an incremental zero-sequence conductance ΔG_0 .

$$\Delta G_0 = \text{Re}\left[\frac{\Delta\bar{I}_0}{\Delta\bar{V}_0}\right] \quad (3.14)$$

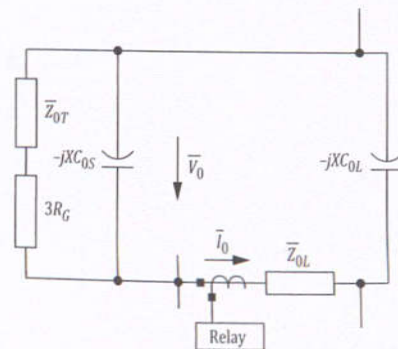
The measured ΔG_0 values coincide with the G_0 values given by (3.12) and (3.13). Incremental quantities eliminate the effect of power system and CT unbalances.

The 32C element compares the ΔG_0 values with forward-fault (negative) and reverse-fault (positive) thresholds. A ΔG_0 value below the negative threshold indicates a forward fault, and a ΔG_0 value above the positive threshold indicates a reverse fault. The 32C element is not affected by system unbalance and could be used with conventional CTs. However, flux-summing CTs are better for improving sensitivity.

3.6.3 High-resistance grounded systems

In resistance-grounded systems, the angle of the local source zero-sequence impedance is different from 90 degrees. With a proper setting of the maximum torque angle $ZOMTA$, you can use the zero-sequence, voltage-polarized directional element (32V) in resistance-grounded systems.

Figure 3.19 shows the zero-sequence network for a forward ground fault in a resistance-grounded system. R_G is the grounding resistance. $\bar{Z}_{0T}$ is the zero-sequence impedance of the distribution transformer or the grounding transformer. The system zero-sequence impedance $\bar{Z}_{0S}$, behind the relay in this case, is the parallel of the series combination of $3R_G$ and $\bar{Z}_{0T}$, with the total capacitive reactance $-jXC_{0S}$ of the remaining system. For this forward fault, the measured zero-sequence impedance equals the equivalent source impedance behind the relay $-\bar{Z}_{0S}$. A similar analysis shows that for reverse faults, the measured impedance



equals the protected line capacitive reactance $-jXC_{0L}$ (assuming that $XC_{0L} \gg Z_{0L}$). If loads are grounded, the impedance measured for reverse faults is the parallel of the load impedance (not shown in Figure 3.19) with $-jXC_{0L}$.

In a high-resistance grounded system, the grounding resistance has a typical value of $3R_G = XC_{0S}$. We can disregard the transformer impedance in this case ($3R_G \gg Z_{0T}$). The system zero-sequence impedance $\bar{Z}_{0S}$ has an angle of -45 degrees and the angle of $-\bar{Z}_{0S}$ is 135 degrees. Therefore, during a forward fault, the zero-sequence current $\bar{I}_0$ of the faulted feeder roughly lags the system zero-sequence voltage $\bar{V}_0$ by 135 degrees, as shown in Figure 3.20(a). The measured zero-sequence impedance $-\bar{Z}_{0S}$ plots on the second quadrant of the impedance plane (Figure 3.20(b)).

You should set $ZOMTA$ between 0 and -90 degrees, for example, $ZOMTA = -45$ degrees. The $32V$ operating characteristic (Figure 3.20(b)) accommodates $-\bar{Z}_{0S}$ values corresponding to different R_G values. Apply flux-summing CTs and sensitive ground elements in high-resistance grounded systems.

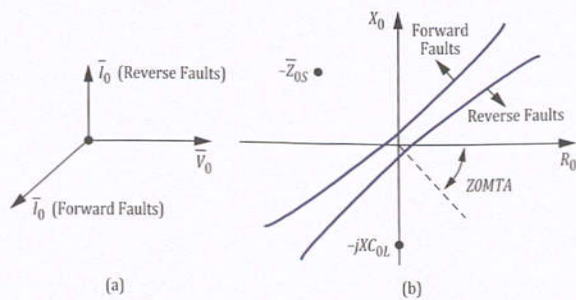


Figure 3.20 Application of a zero-sequence, voltage-polarized directional element (32V) to a high-resistance grounded system.

The wattmetric directional element (32W) and the incremental conductance directional element (32C) are also effective in high-resistance grounded systems. The grounding resistance increases active power consumption during forward faults, a situation that improves the sensitivity of these elements for forward faults.

3.6.4 Effectively and low-impedance grounded systems

Effectively grounded distribution systems are typically four-wire, multigrounded systems. The neutral wire is connected to ground at the substation, at every distribution transformer, and every 1000 feet or so, if there is no transformer ground throughout the feeders.

Low-impedance grounded systems are typically grounded at the substation. A low-value resistor or

reactor is connected to the neutral of the distribution transformer or a grounding transformer.

3.6.4.1 Multigrounded systems

Ground overcurrent elements set above the maximum expected load unbalance are the traditional solution for ground-fault protection of radial distribution systems. However, unbalance limits protection sensitivity.

In multigrounded systems, both load unbalance and ground-fault currents divide between the neutral conductor and earth. Detecting high-impedance ground faults on these systems is difficult because the relay measures the ground-fault current combined with the unbalance current [12].

Fallen power conductors on poorly conductive surfaces, tree branches contacting power lines, and dirty insulators are typical causes of high-impedance faults. The fault current is generally less than 100 A and may equal zero when the conductor falls on asphalt or dry sand. Downed power conductors are a major public safety concern, hazardous to people and property.

Figure 3.21 shows the block diagram of a high-impedance fault detection system that calculates the total nonharmonic content of phase currents to detect the arcing activity on the downed conductor [12], [13]. The Sum of Difference Current (SDI) function calculates the sum of the one-cycle difference phase current over a time interval. The one-cycle difference removes the dc component and all harmonics. Normal distribution loads cause a very low SDI value. The Infinite-Impulse-Response (IIR) Limiting Averager establishes a long-term reference for the SDI quantity. The Trending and Memory function extracts the fault-arcing characteristics and supplies the result to the Decision Logic function, which makes the fault detection decision. The Blocking Conditions function blocks the fault detection for system events other than high-impedance faults. These system events include high-current faults, large voltage changes, and three-phase events.

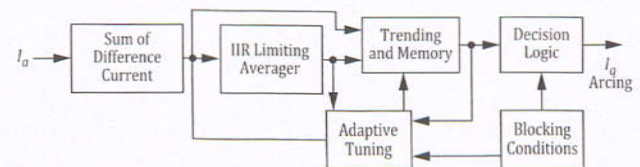


Figure 3.21 A high-impedance fault detection system that uses the total nonharmonic content of phase currents to detect arcing activity.

Nonlinear and dynamic feeder loads may cause current components that resemble arcing activity. The Adaptive Tuning function shown in Figure 3.21 characterizes the feeder load current behavior. Through monitoring, the Adaptive Tuning algorithm learns the current noise profile; it then establishes a comparison threshold for the Trending and Memory function,

increasing the security of high-impedance fault detection.

3.6.4.2 Single-point grounded systems

In single-point grounded distribution systems, ground overcurrent elements measure ground-fault current only, which improves their sensitivity compared to that in multigrounded systems. Applying ground directional elements (Section 3.5.2) further improves ground fault protection sensitivity in single-point grounded systems [14].

We can use Figure 3.19 to analyze the operation of the zero-sequence, voltage-polarized directional element (32V) for forward ground faults in low-resistance grounded systems. Since $3R_G < X_{C0S}$ in these systems, the angle of $\bar{Z}_{0S}$ may be between -45 and 0 degrees, causing the angle of the apparent impedance $-\bar{Z}_{0S}$ to be between 135 and 180 degrees. Furthermore, the zero-sequence reactance $\bar{Z}_{0T}$ of the distribution transformer or grounding transformer may cause the angle of $\bar{Z}_{0S}$ to become positive and the angle of the apparent impedance to be above 180 degrees (third quadrant of the impedance plane). You should set $ZOMTA$ between 0 and -90 degrees. We find the same behavior in low-reactance grounded systems.

Reference [14] provides the method for setting $ZOMTA$ for the 32V element in low-impedance grounded systems.

3.7 Effect of Load Current

3.7.1 Traditional backup sensitivity limitations

In distribution substations (Figure 3.22(a)), traditional bus phase overcurrent elements (51B) must be set above the total load current of all feeders. Therefore, the pickup current of bus phase overcurrent elements (I_{PUB}) is considerably higher than the pickup current of the feeder overcurrent elements (I_{PUF}). In Figure 3.22(b), the feeder fault current I_F is below the bus phase element pickup I_{PUB} ; therefore, the bus relay cannot detect this feeder fault. High pickup settings for traditional bus phase overcurrent elements limit backup coverage for faults at the end feeder sections. They also limit the bus protection sensitivity.

3.7.2 Increasing sensitivity for three-phase faults

Traditional methods for increasing distribution bus relay sensitivity use undervoltage or distance elements to supervise the phase overcurrent elements. Undervoltage supervision is not effective when feeders have fault-current-limiting reactors and the source behind the bus is strong. Under heavy load conditions, the

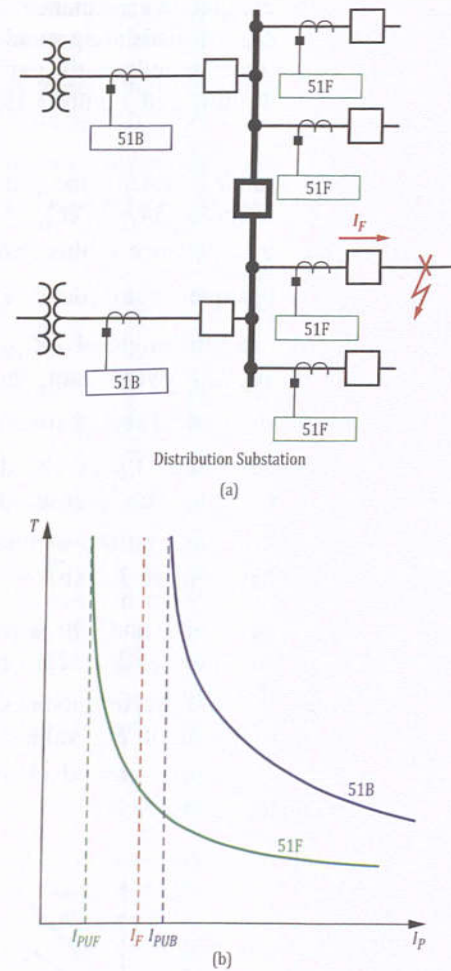


Figure 3.22 High load current makes traditional bus phase overcurrent elements ineffective as feeder relay backup protection.

measured impedance could encroach on the distance element characteristic and cause the scheme to misoperate.

SEL invented a load-encroachment element [15] that blocks the operation of phase overcurrent elements if the apparent impedance falls inside its operating region. The load-encroachment element characteristic fits the load region on the impedance plane and is easy to set using maximum load data. Section 4.6.4 describes the load-encroachment element.

To increase sensitivity for three-phase faults, you can use this load-encroachment element in SEL relays and set the phase overcurrent elements below the bus load current. SEL Application Guide AG2005-07 provides guidelines for applying load-encroachment elements.

3.7.3 Increasing sensitivity for phase-to-phase faults

Negative-sequence overcurrent elements (Section 3.4) provide faster and more sensitive phase-to-phase fault protection. You can set negative-sequence overcurrent elements significantly below the distribution bus load

current. Phase overcurrent elements supervised by a load-encroachment element provide a complementary solution for three-phase faults.

3.7.4 Solving cold-load restoration current problems

Reconnection of a feeder after a prolonged outage causes transient current that remains above the normal load value for seconds or even minutes. High pickup current, the traditional method for increasing overcurrent element security, desensitizes protection.

Programmable logic and group settings in SEL relays solve the cold-load restoration problem. Figure 3.23 shows the cold-load restoration scheme. The Logic and Timing block recognizes a prolonged outage and activates a second group of settings that are tolerant of cold load for a short time after feeder reconnection. The scheme uses the torque control input of phase and ground inverse-time overcurrent elements to modify their pickup current. The breaker trip signal, the recloser lockout signal, and the enable signal for the cold-load restoration scheme are inputs to the Logic and Timing block. Under normal operating conditions, the breaker is closed and the scheme has the normal pickup current I_{PU1} enabled. If the breaker has been open for a given time, the scheme activates the cold-load pickup setting I_{PU2} , as shown in Figure 3.23, without altering the operating time for fault currents. At a set time after the breaker closes to restore service, the scheme restores the normal pickup current I_{PU1} . SEL Application Guides AG99-09, AG2002-22, and AG2002-25 provide guidelines for implementing cold-load restoration schemes with SEL relays.

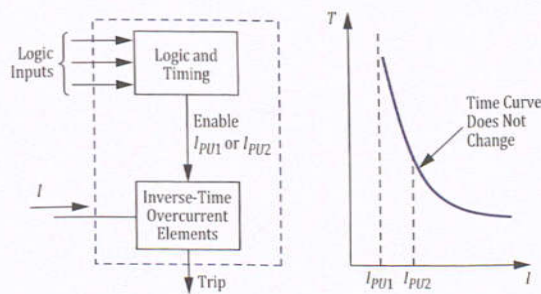


Figure 3.23 Cold-load restoration scheme improves sensitivity by temporarily changing the pickup current of phase and ground overcurrent elements without modifying their time-current curve.

3.7.5 Avoiding sympathetic tripping

In sympathetic tripping, an overcurrent element operates incorrectly if an out-of-section fault causes abnormal load currents. There are two possible sympathetic tripping cases:

- Sympathetic trips can be caused by delayed voltage recovery after an external fault clearance. Tripping may take place many seconds after the initial fault.

The primary cause of these trips is stalled, or stalling, low-inertia air conditioner compressor motors.

- Sympathetic trips can occur during an out-of-section fault. A typical case is a low-set ground overcurrent element tripping for an out-of-section ground fault. The primary cause of these trips is three-phase grounded-wye load or single-phase-to-ground connected load.

Reference [16] proposes two relay logic schemes to avoid sympathetic trips. These schemes detect impending sympathetic trip situations and temporarily increase overcurrent element pickup.

3.8 Distributed Generation Considerations

Increasingly, utility customers are installing distributed generation (DG) in distribution and subtransmission systems. DG uses many forms of energy, such as wind, solar, biomass, hydroelectric, and fossil fuel energy. During major system disturbances and blackouts, DG can provide reliable on-site generation that keeps critical loads in service.

Utilities typically prevent DG sources from feeding a disconnected section of the utility network (an islanding condition) after a utility system outage. When the utility breaker trips, the DG source may continue supporting fault current and may also cause overvoltage in the network. Both overcurrent and overvoltage could damage system equipment and cause a safety concern. Without the utility system operating in parallel, the DG source may not satisfy utility power quality standards and could damage utility customer equipment. In addition, utility and DG equipment may suffer damage if an asynchronous breaker reclosure occurs.

However, customers prefer continuity of service even during islanding conditions. As DG source control systems improve, islanded operation will increasingly support continuity of service. Successful islanded operation requires adaptive load shedding to match DG to islanded load.

3.8.1 Interconnection protection

Interconnection protection disconnects the DG source from the paralleled utility source when the utility source is lost. DG separation prevents system and DG equipment damage caused by fault current, overvoltage, or asynchronous breaker reclosure.

Islanding detection is an important component of the interconnection protection scheme. Islanding detection schemes include:

- Local-area schemes.
- Wide-area schemes.

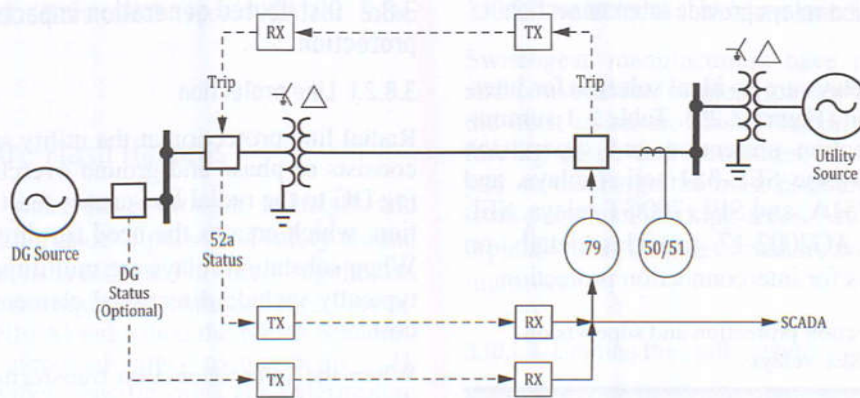


Figure 3.28 Direct transfer trip over a communications channel provides islanding detection when DG active power generation matches the local-area load.

- Directly transfer-trip either the customer main interconnect breaker or the generator breaker(s) when the utility breaker trips.
- Send the status of the main interconnect breaker and/or generator breaker back to the utility substation. This information can confirm that the DG source is isolated before reclosing the substation line breaker.
- Send breaker status information to utility operators via the utility supervisory control and data acquisition (SCADA) system.

Typical communications media between the DG site and the utility substation include fiber-optic cable or low-cost, digital spread-spectrum radios. SEL multi-function relays with secure relay-to-relay communication via MIRRORING BITS® work very well for this application.

System topology-based schemes rely on breaker status, open phase detectors, and trip commands to detect islanding conditions and isolate the DG. These schemes are very simple in concept, but they need to adapt to topology changes in the power system. This requirement can result in a system with many communications links and poor reliability.

Section 8.11 describes a synchrophasor-based wide-area scheme that uses a synchrophasor vector processor (SVP; Section 2.6.2) to process synchrophasor information measured by relays installed at the DG location and a remote location. These relays send synchrophasor voltage messages to the SVP through a communications channel. The SVP calculates the angle difference between these voltages, the rate of change of the angle difference (slip frequency), and the rate of change of the slip frequency (angle acceleration) as described in Section 8.10.1. The SVP declares an islanding condition when the angle difference is greater than a threshold value or when an out-of-step tripping element (Section 8.10.3), running on the SVP, declares an out-of-step condition. This scheme reliably detects islanding conditions even when the DG active power genera-

tion is close to the islanded area load because the islanded system voltage eventually slips away from the utility system voltage. Scheme operation does not depend on system topology.

3.8.1.3 Complete interconnection protection

An interconnection protection scheme for a DG installation includes islanding detection, external fault protection, and additional equipment protection elements. Reference [17] presents a typical interconnection protection scheme.

Phase directional overcurrent elements (67) supervised by load-encroachment elements provide external three-phase fault protection. Load-encroachment supervision allows the use of sensitive settings in directional overcurrent elements and provides security for normal load conditions.

Negative-sequence directional overcurrent elements detect phase-to-phase and phase-to-ground faults on the high side of the delta-wye transformer.

Reference [8] analyzes the application of directional elements at the utility-industrial interface and provides application guidelines and settings recommendations.

The method for detecting external ground faults depends on how the interconnection transformer is connected. When the transformer has a delta connection on the utility (high-voltage) side (as in Figure 3.28), apply voltage elements measuring the high-side voltage [17]. When the transformer has a grounded-wye connection on the utility side, use a ground overcurrent element measuring the transformer neutral current (51G).

Phase current unbalance (46) and/or phase voltage unbalance (47) elements prevent equipment damage. Conduct transient stability studies to determine whether the DG source requires out-of-step protection (78).

3.8.1.4 SEL multifunction relays provide interconnection protection

SEL multifunction relays are an ideal solution for interconnection protection (Figure 3.29). Table 3.1 summarizes the interconnection protection and supervision elements provided by the SEL-351 series relays, and the SEL-547, SEL-751A, and SEL-700GT relays. SEL Application Guide AG2002-17 provides details on using SEL-351 relays for interconnection protection.

Table 3.1 Interconnection protection and supervision elements provided by SEL relays.

Element	Relay
Over-/Undervoltage Over-/Underfrequency Directional power Synchronism checking	SEL-547 SEL-751A
Over-/Undervoltage Over-/Underfrequency Directional overcurrent Synchronism checking	SEL-351A
Over-/Undervoltage Over-/Underfrequency Directional power Directional overcurrent Synchronism checking	SEL-351-7 SEL-351S
Over-/Undervoltage Over-/Underfrequency Frequency rate of change Directional power Directional overcurrent Out-of-step protection Synchronism checking Automatic synchronization	SEL-700GT



Figure 3.29 SEL-351 relays protect a wind farm. Multifunction SEL relays provide comprehensive interconnection protection for DG installations.

3.8.2 Distributed generation impacts utility system protection

3.8.2.1 Line protection

Radial line protection at the utility substation typically consists of phase and ground overcurrent relays. Adding DG to the radial line causes fault current redistribution, which creates the need for directional protection. When substation relays are multifunction devices, they typically include directional elements at no additional cost.

When the interconnection transformer has a high-side delta winding, there is no zero-sequence current contribution to external ground faults. Therefore, adding a DG source with such an interconnection transformer to the line does not significantly affect substation ground overcurrent relays.

3.8.2.2 Automatic reclosing and synchronism checking

Automatic reclosing time delays at the utility substation should accommodate islanding scheme tripping times. Users should apply voltage or synchronism-checking functions to supervise automatic or manual reclosing at the substation. A communications channel, if available, can confirm that generation is offline before permitting the breaker to close [17].

3.9 High-Speed Distribution System Protection

Fast fault clearing preserves system stability, reduces arc-flash hazards and equipment damage, and improves power quality. In the past, high-speed tripping was only considered necessary for transmission systems. Now, modern distribution and subtransmission systems also require high-speed protection [19], because loads are becoming more sensitive to voltage sags caused by faults.

Delayed fault clearing may cause voltage stability problems. For example, in areas with a large air conditioning load, the voltage sag caused by a fault can initiate a voltage collapse. Distribution or subtransmission systems with DG also require high-speed tripping to preserve transient stability. Typical fault-clearing times of traditional overcurrent protection are more than 0.5 seconds, which is unacceptable for modern systems.

The communication built into today's relays provides high-speed, communication-based schemes. Section 6.4.5 describes low-cost, zone-interlocked protection schemes that provide fast bus fault clearing using transformer and feeder relays. References [20], [21], and [22] describe high-speed distribution and subtransmission line protection systems using low-cost spread-spectrum radio and fiber-optic channels. SEL multifunction relays with secure relay-to-relay communica-

tion via MIRRORING BITS work very well for these applications.

3.10 Reducing Arc-Flash Hazards

The past several years have brought a heightened awareness of the importance of personnel safety around electrical apparatus, as is reflected in recent regulations and standards. For example, the National Fire Prevention Association (NFPA) published the NFPA Standard 70E to document electrical safety requirements [23]. This standard provides specific rules for determining the category of electrical hazards and the personal protective equipment required in the defined and marked hazard zones.

In particular, industry and utilities recognize that arc-flash events can cause dangerous and potentially fatal levels of heat, ultraviolet radiation, blast pressure, flying shrapnel, and deafening sound waves. The existing standards are mainly concerned with the heat energy from the arc flash. IEEE Standard 1584 [24], for example, provides information on how to calculate arc energy and establish boundary distances for personnel when working around energized electrical equipment. References [25] and [26] provide detailed arc-flash calculation examples.

3.10.1 Methods for reducing arc-flash hazards

The energy produced by an arc-flash event is proportional to the voltage, current, and duration of the event. Design engineers can reduce system voltage or fault currents, but the best and most direct ways to reduce arc-flash hazards are to prevent personnel from working in danger zones and to shorten fault-clearing times.

The most common methods for reducing arc-flash hazards are [26]:

- Avoiding the hazard area.
- Installing arc-resistant switchgear.
- Limiting the fault current.
- Improving protection schemes.

3.10.1.1 Avoiding the hazard area

The safest way to prevent arc-flash injuries is to avoid the danger zone. Performing work on de-energized equipment eliminates working in hazard areas. Technology offers several ways to gather information and perform operations without entering the hazard area [27]. Communications links to the equipment provide key maintenance and operating data. Manufacturers provide remotely controlled breaker racking mechanisms to perform actions away from the arc-flash zone.

3.10.1.2 Installing arc-resistant switchgear

Switchgear manufacturers have modified the design and construction of electrical switchgear to withstand the blast of an arc flash. Enhancements include reinforcing doors and structures, as well as providing a path to discharge the blast pressure and material away from personnel working areas. Arc-resistant switchgear typically also includes breakers with high-speed clearing times.

3.10.1.3 Limiting the fault current

Reducing the available fault current lowers the arc energy, thus limiting the hazard. High-impedance transformers and series reactors reduce phase and ground-fault current, and impedance grounding reduces the ground-fault current. On low- and medium-voltage systems, current-limiting fuses provide fast clearing times and limit the fault current.

3.10.1.4 Improving protection schemes

The new abilities in microprocessor-based relays provide many low-cost ways to reduce arc-flash hazard by reducing fault-clearing times. Possible improvements to distribution protection schemes include:

- Faster overcurrent protection:
 - Lower coordination time intervals.
 - Applying instantaneous elements.
 - Applying negative-sequence overcurrent elements (Section 3.4).
- Adding bus differential protection (Section 6.4.1, Section 6.4.2, and Section 6.4.3).
- Adding zone-interlocked protection (Section 6.4.5).
- Adding breaker-failure protection (Section 6.5).
- Adding high-speed line protection with fiber-optic or radio channels (Section 3.9).
- Enabling an instantaneous overcurrent element during maintenance.
- Adding arc-flash protection.

Enabling an instantaneous overcurrent element during maintenance improves safety for personnel working near energized equipment. You can enable the instantaneous element using a pushbutton on the relay, a separate switch, or a communications channel. While activated, this scheme disables the time coordination and allows the breaker to trip without any delay. Activate this special protection scheme when workers are near the energized switchgear. You can add this scheme to new or old installations at low cost.

3.10.2 Arc-flash protection

Arc-flash protection uses fiber-optic light sensors located in metal-enclosed and metal-clad switchgear to detect high-energy arc-flash events and initiate high-

speed breaker tripping. Including arc-flash detection in a protective relay, such as the SEL-751A, minimizes cost and complexity by using the relay hardware and software and the existing tripping circuits. The added advantage of processing arc-flash detection in the relay is the ability to use a fast overcurrent element as a supervising element to improve security. The SEL-751A provides four arc-flash protection elements. Each element includes:

- Arc-flash time-overlight™ (TOL) element.
- Arc-flash overcurrent element.

3.10.2.1 Arc-flash time-overlight element

Arc-flash detection sensors provide a clear measurement of an arc flash. The light emitted during an arc-flash event is significantly brighter than normal ambient light. The light surge is noticeable as soon as the flash starts and is easy to detect using proven technology. The SEL-751A relay supports four fiber-optic light sensors of two distinct types: point sensors and bare-fiber sensors. Fiber-optic cables bring the light signals from the sensors to the detectors located in the relay. Figure 3.30 shows a rear view of the SEL-751A relay with a point sensor connected to one of the four inputs available in the optional arc-flash card.

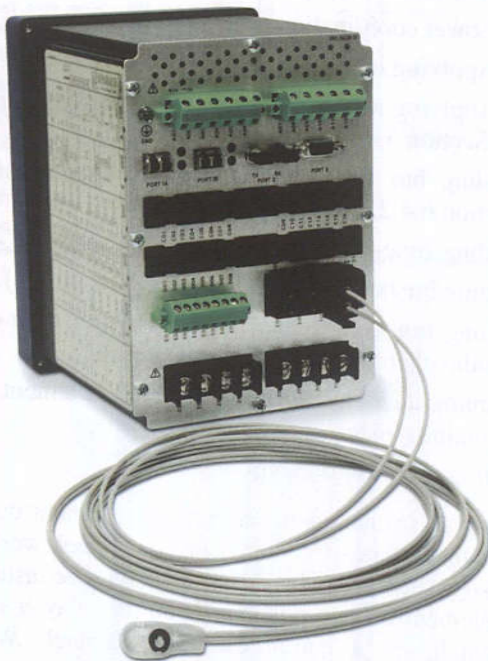


Figure 3.30 SEL-751A with a point sensor connected to the arc-flash card.

The point sensor is omnidirectional and is optimized for installation in individual switchgear compartments. The bare-fiber sensor is a high-quality plastic fiber-optic cable without a jacket (1,000 μm). This clear fiber-optic cable acts as a lens, bringing in light from

the area. The bare-fiber sensor is optimized for arc-flash detection in long, distributed resources, such as the switchgear bus compartment.

Using a fiber-optic loop connection allows the relay to test the system by periodically injecting light. This loop testing system works with either the point sensor or the bare-fiber sensor. Each sensor has one end connected to a transmitter in the relay and the other end connected to a detector in the relay. The relay continually monitors all arc-flash sensors by pulsing them with light every 10 minutes. This diagnostic system asserts a Relay Word bit if a sensor is damaged.

The operating time of the TOL elements in the SEL-751A relay is an inverse function of the input light level, expressed in multiples of the TOL element pickup value (Figure 3.31). The shape of the inverse-time characteristic is fixed to provide robust rejection of unrelated light events without adding unnecessary settings. This characteristic offers enhanced security and exceptionally fast operation.

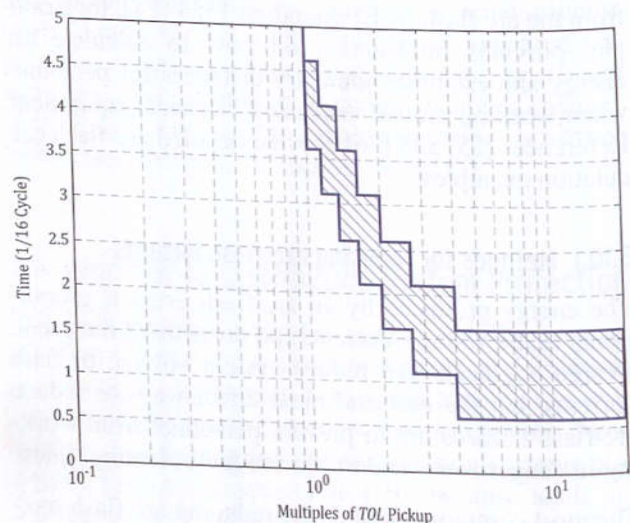


Figure 3.31 Inverse-time characteristic of the SEL-751A arc-flash TOL elements.

Given the critical nature of the arc-flash protection function, you should set the TOL pickup value based on the ambient light level. This approach guarantees maximum sensitivity and provides the fastest tripping time.

The SEL-751A relay continuously measures ambient light. You can display its value using the front-panel **METER > Light Intensity** menu, as well as the **MET L** command. Set the TOL pickup value above the highest normal light intensity expected at each arc-flash sensor installation. Arc-flash events significantly exceed virtually all illumination levels normally found in a switchgear environment. The only exception is exposure to direct sunlight, which can reach or exceed arc-flash TOL element setting thresholds.

3.10.2.2 Arc-flash overcurrent element

Supervising the TOL element with a high-speed overcurrent element improves security without sacrificing tripping speeds [26]. Many standard overcurrent elements have operating times between 6 and 20 milliseconds. These elements are slower than the TOL element. To avoid introducing additional delay, the high-speed overcurrent element must be as fast as the arc-flash detection.

The SEL-751A relay includes one three-phase and one ground instantaneous overcurrent element for arc-flash detection supervision. These arc-flash overcurrent elements use raw analog-to-digital converter samples, with a sampling rate of 16 samples per cycle. Individual samples are compared with the setting threshold, followed by a security counter that requires two samples in a row to be above the setting threshold.

SEL-751A arc-flash protection is exceptionally fast. Typical relay operating times are approximately 2–5 milliseconds when equipped with the optional fast-hybrid output card. With standard, electromechanical outputs, tripping time increases to 7–13 milliseconds.

The arc-flash overcurrent elements do not reject harmonics and therefore have a natural tendency to operate under high harmonic load conditions. To avoid unintended element pickup, set the arc-flash overcurrent element pickup at least two times the expected maximum load. TOL element supervision will prevent temporary activation of the arc-flash overcurrent element during cold-load pickup conditions from causing misoperation.

3.10.2.3 Event reports

Relay event reports show light and current analog measurements (Figure 3.32). By monitoring the incoming light as an analog signal, users can view and set the normal light levels for the application. The event reporting also provides a troubleshooting tool with time-tagged events, including sensor light levels. Tracking the arc-flash intensity provides the detail needed to determine the root cause of an event.

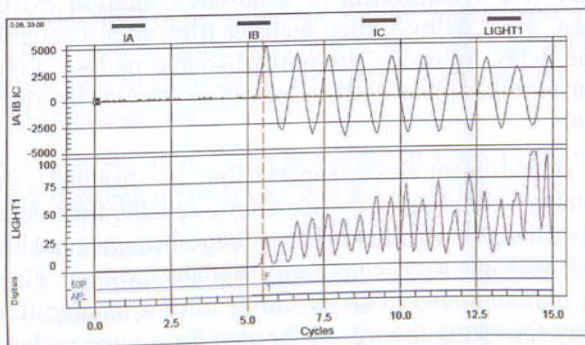


Figure 3.32 SEL-751A fault event report containing current and light data.

3.10.2.4 Sensor location

Arc-flash detection systems typically use a combination of point sensors and bare-fiber sensors connected to a single relay. Locate sensors where arc-flash detection for the specific sensor would trip the corresponding upstream breaker. Installation of sensors varies, depending on the switchgear manufacturer, type of gear, and number of sections. Multiple sensor inputs in the relay provide coverage and sectioning options.

Figure 3.33 shows a typical application example in switchgear with one incoming and two radial (outgoing) feeders. SEL-751A relays controlling breakers 1, 2, and 3 protect all three feeders. Feeder breakers 2 and 3 must trip for downstream faults, normally located in the outgoing cable termination compartment. To obtain better coverage, install multiple sensors in the same compartment, as shown in Figure 3.33, with point sensors LS1 and LS2 connected to Relay 2.

Relay 1 and Breaker 1 protect the bus compartment and the outgoing breaker compartments for breakers 2 and 3. Sensors LS1, LS2, and LS3 connect directly to Relay 1. Sensor LS1 is a bare-fiber loop enclosing the entire length of the bus.

You can also use radial feeder relay sensors (such as LS3 connected to Relay 2) to transfer-trip the upstream breaker. Implement this logic using the relay SELOGIC® programming ability.

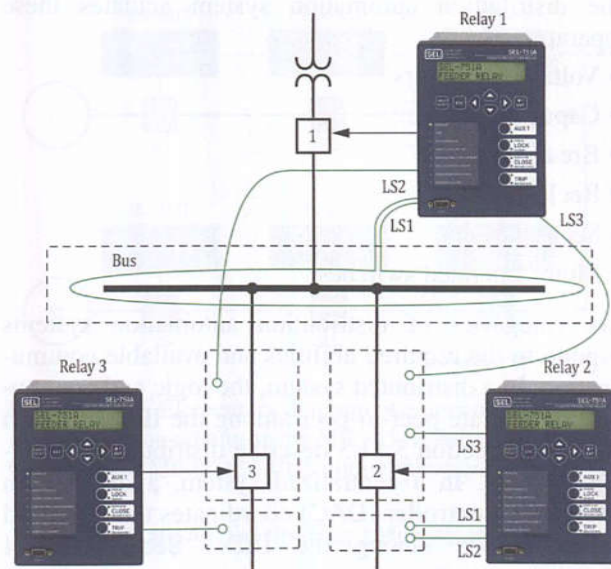


Figure 3.33 Switchgear application example.

3.11 Distribution Automation

3.11.1 Distribution automation objectives

Modern distribution automation (an important element of smart grid technology) uses intelligent electronic devices (IEDs) and communications networks to auto-

matically and rapidly isolate faults, restore power, and control reliable and economical operation of the distribution system. Multifunction relays are an economical solution for distribution automation because they measure system voltages and currents, monitor the status of breakers, reclosers, and switches, communicate between devices, and are programmable.

The objectives of distribution automation include:

- Optimizing distribution system operation (minimizing power losses and correcting power factor).
- Controlling voltage.
- Load balancing and load shedding.
- Improving service availability.
- Improving service restoration.
- Enhancing power quality.
- Accommodating distributed generation sources.
- Enhancing situational awareness.

These objectives are interdependent [28]. For example, restoring service to a feeder section may result in overload conditions, excessive voltage drop, or power factor degradation. The control system may need to transfer load from the overloaded feeder to adjacent feeders or perhaps shed noncritical load to alleviate actual or predicted overloads. The control system may also need to adjust voltage regulators to alleviate voltage drops or switch capacitors to correct power factor and reduce losses in the new feeder configuration.

The distribution automation system actuates these apparatus:

- Voltage regulators.
- Capacitor banks.
- Breakers.
- Reclosers.
- Sectionalizers.
- Motor-operated switches.

Two categories of distribution automation systems respond to the required abilities and available communication. In a distributed system, the logic and communication operate peer-to-peer among the IEDs. Section 3.11.2 and Section 3.11.3 describe distributed automation systems. In a centralized system, a distribution automation controller (DAC) coordinates the logic and communication among the IEDs. Section 3.11.4 describes centralized automation systems.

3.11.2 Automatic throw-over schemes

Throw-over schemes monitor the line voltages from two sources and automatically transfer the load to a healthy source during service interruptions. SEL Application Guide AG2002-23 provides general guidelines for implementing an automatic throw-over scheme

using SEL-351R Recloser Controls and MIRRORED BITS communications.

Figure 3.34 depicts the throw-over scheme, which may use different communications channels, such as fiber-optic cable, digital spread-spectrum radio, copper wire, or telephone line. Both recloser controls provide standard overcurrent protection and reclosing functions. At each location, the recloser control monitors the recloser status and the voltages at each side of the recloser. Each control has communications ports suitable for communicating between the two controls over different channels.

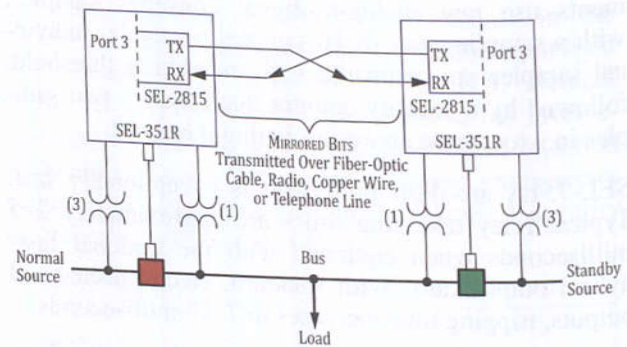


Figure 3.34 Automatic throw-over scheme using SEL-351R Recloser Controls and MIRRORED BITS communications.

Under normal conditions, the load is fed through the normal-source recloser, and the standby-source recloser is open. Both recloser controls continuously monitor phase voltage magnitudes. The normal-source recloser also provides overcurrent protection.

When any phase voltage measured by the normal-source recloser control drops below a preset level and an overcurrent condition does not exist, the normal-source recloser trips after a settable time delay. If the normal-source recloser trips on a low-voltage condition, the standby-source recloser closes after a settable time delay and restores service to the load.

When the normal-source voltage returns to a healthy state and synchronism-close supervision is enabled, the normal-source recloser closes after a settable time delay if a synchronism or dead-bus condition exists. Then, the standby-source recloser trips after a settable time delay following the normal-source reclose. Load transfer takes place without temporary service interruption.

If synchronism-close supervision is disabled, the standby-source recloser trips after a settable time delay when the normal-source line voltage becomes healthy (the load again is temporarily without power). Then, the normal-source recloser closes after a settable time delay following the trip of the standby-source recloser to restore power to the load.

Reference [29] provides application examples of the automatic throw-over scheme (Figure 3.34). A scheme

installed at a large industrial system uses fiber-optic communication between two recloser controls separated by one pole span on a distribution circuit. Another scheme uses digital point-to-point radios. Yet another throw-over scheme uses a pad-mounted transfer switch instead of reclosers.

3.11.3 Distribution network fast-restoration schemes

The effect of a service interruption depends on the type of load. Reference [30] summarizes a survey of critical service loss durations for industrial plants. Twenty-five percent of industrial plants must completely restart production if service is interrupted for more than 10 cycles. The average time to restart these industrial plants is more than 17 hours. Even short service interruptions may cause significant economic loss to the customer.

Many utilities still restore distribution loads after a fault by manually operating switches and other sectionalizers to isolate the faulted line. One way of reducing restoration times is to apply microprocessor-based relays and/or recloser controls at each switch location with voltage signals supplied from voltage transformers at each side of the switch. The devices detect hot/dead voltage conditions, thereby allowing fast automatic tripping and restoration of switches. Implementing this scheme with SEL-351R or SEL-651R Recloser Controls reduces restoration times of nonfaulted lines from minutes (manual restoration) to seconds.

Using intelligent relays or recloser controls linked together by communications channels can further reduce load restoration times from minutes or seconds to cycles. SEL Application Guide AG2004-09 provides general guidelines for using SEL-651R Advanced Recloser Controls and MIRRORRED BITS communications to implement high-speed reconfiguration schemes in distribution networks.

Figure 3.35 shows the elementary diagram of a network reconfiguration scheme. Each SEL-651R communicates with adjacent recloser controls via a MIRRORRED BITS communications channel. Typical communications media between recloser controls include fiber-optic cable or digital spread-spectrum radios. This scheme does not require voltage inputs, other than those used to power the relay and provide trip and close energy.

An example of the system operation for a permanent fault between reclosers R2 and R3 is as follows:

1. Instantaneous overcurrent elements in the R1 and R2 recloser controls pick up for the fault. A timer with delayed dropout asserts in each control. The overcurrent element in the R3 recloser control does not pick up because the system is radial (R4 is open).

2. Recloser R2 operates to lock out for the fault. The delayed-dropout timer in R1 remains asserted for a time period after the fault has cleared.
3. Once R2 has tripped and locked out for the fault, the SEL-651R controlling R2 sends a MIRRORRED BIT to the R1 and R3 recloser controls.
4. When the R1 recloser control receives the bit, no action occurs because R1 tripping is supervised by the delayed-dropout timer.
5. When the R3 recloser control receives the bit, R3 trips and is locked out because the overcurrent element did not assert the timer at this location. At this point, the faulted line section (R2–R3) has been isolated; however, the unfaulted line section between R3 and the normally open R4 recloser is also out.
6. Once R3 has opened, the R3 recloser control sends a MIRRORRED BIT to the R4 recloser control.
7. When the R4 recloser control receives the bit, R4 closes to restore the unfaulted line section.
8. Following this operation sequence, the scheme turns itself off to avoid scheme action for additional faults that may occur.

Reference [31] describes a practical application of the high-speed distribution network reconfiguration scheme depicted in Figure 3.35.

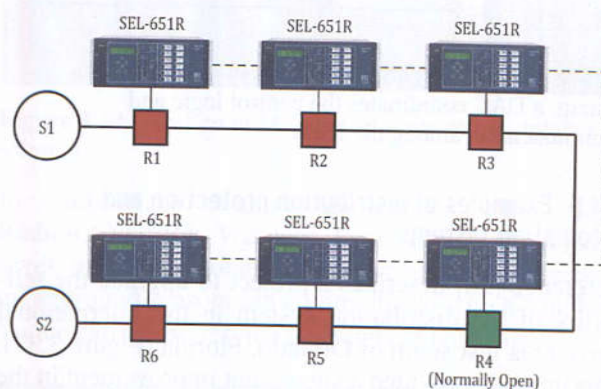


Figure 3.35 Elementary diagram of a network reconfiguration scheme using SEL-651R Advanced Recloser Controls and MIRRORRED BITS communications.

3.11.4 Centralized distribution automation systems

The schemes we have described apply to simple distribution networks. Controlling networks with many feeders and complex configurations requires sharing more information via communications channels. A centralized system, based on a DAC, is a good solution for complex networks. The DAC collects information from the IEDs and apparatus to monitor network topology, feeder loading, voltage levels, and other valuable information throughout the network. The DAC makes

system-wide control decisions to achieve the automation objectives listed in Section 3.11.1.

You can use the SEL-2440 Discrete Programmable Automation Controller (DPAC) or the SEL-3530 Real-Time Automation Controller (RTAC) as the DAC of the centralized automation system. The distribution substation is often a convenient location for the DAC. However, the DAC can be located anywhere the necessary communications channels are available.

The centralized system may have a star configuration (Figure 10.5) or a ring (looped) configuration, as shown in Figure 3.36. The DAC acts as the controller for the distribution automation system and the communications network. A centralized system described in [32] uses meshed radios and multidrop communication to collect information from all the relays and recloser controls, and to send control commands to different network locations.

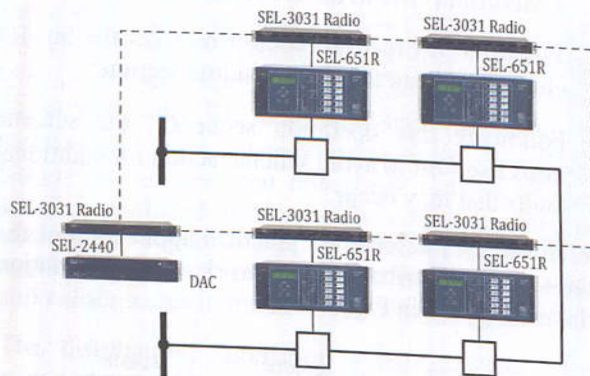


Figure 3.36 In a centralized distribution automation system, a DAC coordinates the control logic and communication among the IEDs.

3.11.5 Examples of distribution protection and automation systems

Reference [20] describes a project to upgrade the reliability of the distribution system in the International Drive area just south of Orlando, Florida (Figure 3.37). This upgrade provided a significant improvement in the protection, control, and automation of the system.

The underground system operates as a closed loop. Directional comparison schemes provide high-speed line protection. These communication-assisted schemes use either permissive overreaching transfer trip logic or a blocking logic. Other solutions for reducing fault-clearing times include fast bus-tripping and breaker-failure protection schemes.

The relays also perform an automatic source transfer. This scheme protects against the loss of a substation bus at Orangewood Substation, which has two buses with a normally open tie breaker. One bus supplies Loop 1, the other bus, Loop 2. If the system loses one

transformer at the substation, the tie breaker automatically closes to restore service to both loops. However, in the case of a bus fault, the bus protection scheme operates to clear the fault, and the loop supplied by this bus section loses service. In this case, the switchgear connecting to the next loop detects the loss of source on both of its two normally closed feeder positions. After a time delay to coordinate with the substation throw-over, the tie-feeder closes, provided voltage is present on the alternate source.

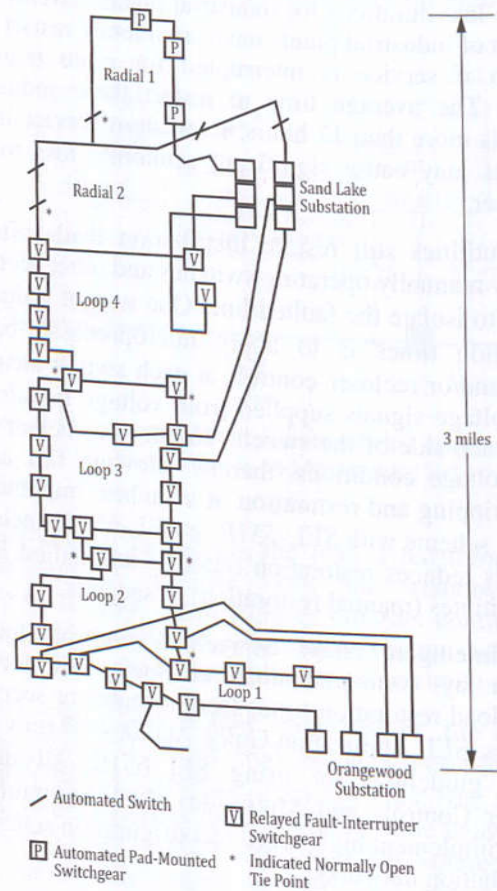


Figure 3.37 International Drive system single-line diagram.

Reference [33] describes another example scheme, a fast-restoration scheme for loss of primary-source interconnection of a distribution system. Reference [32] describes distribution automation systems used by three utilities: a distributed fast-restoration scheme and two centralized systems. In one of the centralized systems, the IEDs are installed in a prefabricated, integrated control enclosure. References [34] and [35] describe load-shedding schemes for large industrial power systems. Reference [28] outlines the benefits and strategies of distribution network automation and discusses the impact of automation on distribution network protection and operation.

3.12 Faulted Circuit Indicators

3.12.1 Benefits of faulted circuit indicators

Manually opening, then reclosing, circuit sections is a traditional method for identifying the faulted line in distribution systems. This method is time consuming and potentially dangerous; it repeatedly exposes distribution equipment to fault current.

Faulted circuit indicators (FCIs) improve the fault-locating process. These devices show whether fault current was present at their monitored locations, providing an indication of the fault path. FCIs provide a safe, economical, and effective way to identify the faulted line section.

FCIs help reduce the time needed for fault location and system reconfiguration. Therefore, they help reduce overall outage time and improve reliability metrics, especially when applied as part of a complete distribution system solution.

Today, investor-owned utilities, municipalities, and rural electric cooperatives make up a large percentage of FCI users. The number of applications in industrial and commercial areas is also growing.

3.12.2 Faulted circuit indicator applications

Overhead FCIs are typically single-phase devices with an integral display that communicates its status from the conductor to the ground. Directional FCIs display fault information based on the direction of power flow during a fault.

Overhead FCIs range from simple manual reset devices to microprocessor-based products with sophisticated tripping and resetting methods. They usually have inrush restraint to prevent false indication caused by inrush currents during recloser operations. The SEL AutoRANGER® overhead FCI automatically adjusts its pickup value based on system load (Figure 3.38).

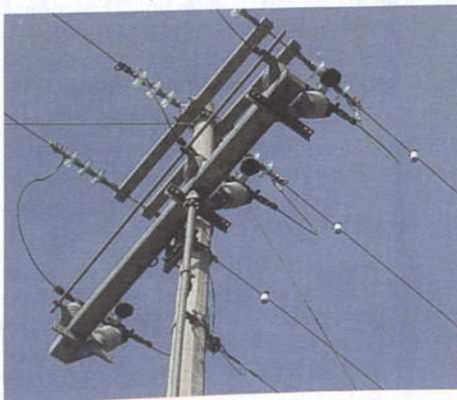


Figure 3.38 Overhead AutoRANGER FCIs automatically adjust to varying load conditions.

Overhead FCIs are valuable at the following locations:

- Along the feeder and at unfused radial taps.
- At the first overhead section as the feeder leaves the substation near the underground-to-overhead transition.
- At each overhead section of feeders that have overhead and underground sections.

Underground FCIs are available in single- or three-phase versions with a variety of integral or remote display options (Figure 3.39). Many FCIs include inrush restraint, auxiliary contacts for integration with a SCADA system, and protection from system backfeed. The SEL Underground AutoRANGER FCI prevents false resetting from backfeed conditions by using historic load values to set the minimum load current reset value.



Figure 3.39 Test point FCIs are an economical, durable solution.

In subsurface vault applications, the SEL RadioRANGER® Wireless Fault Indication System saves time and money and improves safety, because underground sensors can communicate with a handheld device carried by service personnel (Figure 3.40).

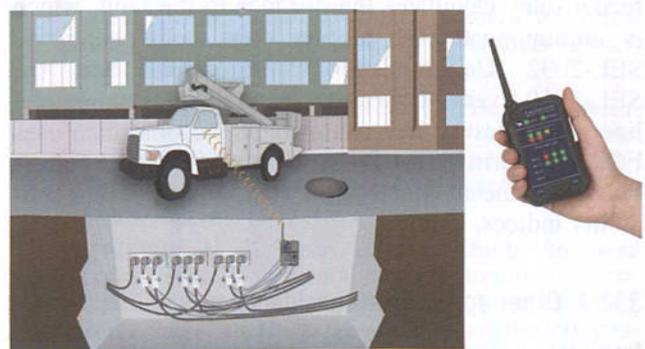


Figure 3.40 The RadioRANGER Wireless Fault Indication System communicates FCI status from phase sensors in an underground vault to a handheld remote fault reader.

Applying underground FCIs with remote displays avoids the problems of opening an enclosure to determine whether an FCI has tripped (Figure 3.41). The displays are visible from the outside of the enclosure and frequently from the line truck. Available remote displays include reflective targets, LEDs, LED and target combinations, acoustical annunciation, SCADA-compatible auxiliary contacts, and radio-frequency links.

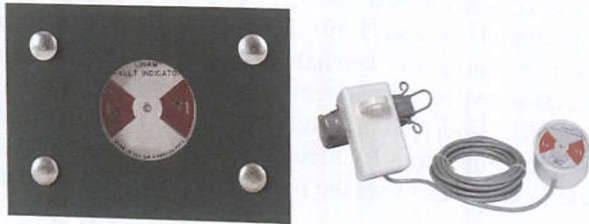


Figure 3.41 Remote displays provide indication from the outside of a pad-mounted transformer, switchgear, and other enclosures to facilitate quick, easy fault locating.

Other possible underground FCI applications include:

- On live-front switchgear, single- and three-phase transformers, and sectionalizing cabinets.
- On underground cables below the stress relief point.

3.12.3 Combine faulted circuit indicators and relays for fast fault location

Modern distribution relays and recloser controls (such as SEL-351, SEL-351S, SEL-451, SEL-751A, SEL-351R, and SEL-651R) have fault location ability. Substation relays determine the fault location as the distance from the substation to the fault. Because most feeders have multiple taps, the relay distance-to-fault information indicates multiple possible fault locations. Therefore, the line crew cannot determine which tap to follow to find the fault.

Combining fault location information from the SEL relays and fault-path information from FCIs promptly finds the faulted section. When a fault occurs, the SEL feeder relay calculates the distance to the fault, which is communicated to a line crew cell phone via SEL-2032 Communications Processors and the SEL-3010 Event Messenger. With this information, the line crew goes to the general fault location, then uses FCIs to determine the faulted section. This technology expedites line repair, reduces cost, and improves reliability indices.

3.12.4 Other application considerations

Impedance grounding limits ground-fault current. FCIs that measure the phase currents have limited sensitivity in impedance-grounded systems.

You can use ground fault detectors (GFDs) in impedance-grounded systems. These devices measure the

zero-sequence current by using a split-core current sensor that surrounds all three phase conductors and can be installed without opening the primary circuit. The GFD trips when the zero-sequence current exceeds a threshold.

3.12.5 Looking to the future

As FCIs adapt to a variety of communications schemes, they will become distribution automation sensors and offer expanded functions beyond fault indication. Soon, FCIs will provide valuable system data to utility control centers. These increasingly intelligent devices will integrate into other system components and become part of the infrastructure necessary to operate modern power systems. The challenges will include preventing data overload at the control centers and managing internal energy at the sensor level.

3.13 References

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